Deep Virtual Meson Production (J/ψ) Analysis Note

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Abstract

The Electron–Ion Collider (EIC) is a next-generation facility to probe the internal structure of nucleons and nuclei via high-luminosity, high-energy electron–hadron collisions. Operating at center-of-mass energies of about 28–140 GeV and luminosities of 10^{33}–10^{34} cm^{-2}s^{-1}, it will enable detailed studies of quark and gluon dynamics. The EIC physics program and detector requirements, documented in the EIC White Paper [1] and Yellow Report [2], led to the design of the general-purpose electron-proton ion collider (ePIC) detector, optimized for excellent tracking, calorimetry, and near-complete angular coverage.

Deeply Virtual Meson Production (DVMP) constitutes a central component of the EIC physics program, as it provides direct access to the three-dimensional internal structure of the nucleon encoded in Generalized Parton Distributions (GPDs). In particular, exclusive vector meson production, such as J/ψ electroproduction, exhibits a pronounced sensitivity to gluon GPDs, thereby serving as a robust probe of the spatial distribution of gluons in both nucleons and nuclei. At sufficiently large photon virtuality and meson mass, exclusive J/ψ production is dominated by gluon-mediated interactions, enabling relatively clean access to gluon imaging and to potential gluon saturation phenomena.

The present study is within the Early Science program of the EIC and focuses on exclusive J/ψ production in electron–proton collisions for a beam configuration of 10 GeV electrons colliding with 130 GeV protons. A comprehensive, simulation-based analysis is carried out employing the lAger event generator [3] with the ePIC detector framework. The exclusive decay channel $J/\psi \to e^+e^-$ is reconstructed, and the detector performance is quantified in terms of kinematic coverage, geometrical and kinematic acceptance, as well as momentum and angular resolutions.

The analysis addresses key observables to GPD studies, including the squared four-momentum transfer t, transverse momentum distributions, and reconstructed kinematic variables characterizing the exclusive process.

Contents

1 Introduction and Motivation

1.1 Introduction

A primary objective of the EIC physics program is to understand the three-dimensional (3D) internal structure of nucleons and nuclei in terms of their quark and gluon degrees of freedom. This information is hidden in GPDs, which generalize conventional parton distribution functions by incorporating the transverse spatial distribution of partons through their dependence on the squared four-momentum transfer, t. Experimentally, GPDs can be probed in hard exclusive processes, most notably Deeply Virtual Compton Scattering (DVCS) and Deeply Virtual Meson Production (DVMP).

Among possible channels from electron-proton interactions, exclusive vector meson production plays a unique role. In particular, exclusive J/ψ production is dominantly sensitive to gluon GPDs due to the large charm-quark mass, which provides a hard scale and suppresses quark-exchange contributions. As a result, the momentum transfer dependence of the exclusive J/ψ cross section enables direct access to the transverse spatial distribution of gluons in the proton, offering a powerful tool for gluon tomography at the EIC.

This analysis focuses on the exclusive process

$$e + p \to e' + p' + J/\psi,$$

with the J/ψ reconstructed via its dilepton decay channel, $J/\psi \to e^+e^-$. The dilepton final state provides a clean experimental signature with excellent mass resolution and well-controlled backgrounds in the ePIC detector. The reconstruction of the scattered electron, decay leptons, and forward proton is essential to establish event exclusivity and accurately determine kinematic observables such as Q^2, x, and t.

The EIC, with its high luminosity, polarized electron and proton beams, and broad kinematic coverage ($\sqrt{s} \approx 28$–140 GeV for ep collisions), is uniquely suited for measurements that probe the internal structure of the proton. In particular, the EIC will extend exclusive J/ψ studies into the small-x and high-Q^2 regime, where gluon dynamics dominate and where sensitivity to non-linear QCD effects may emerge [4, 5].

This work discusses the early scientific opportunities associated with the exclusive physics program foreseen during Phase I of the EIC. In particular, this note reports a simulation-based study of exclusive J/ψ production in the context of the EIC Early Science program. The analysis is carried out using the lAger event generator [3], processed through a detailed model of the ePIC detector. The detector performance is characterized in terms of geometric acceptance, momentum and angular resolution, and reconstruction efficiency, with special attention to the far-forward region ($\eta > 4.5$) that is critical for the detection of the recoil proton. The results provide a realistic evaluation of the EIC's capability to perform precision measurements of exclusive J/ψ production and to constrain the gluon GPDs in the proton.

1.2 Motivation

Studying exclusive J/ψ production at the EIC is a route to extract the transverse spatial distribution of gluons within the proton. Different from inclusive measurements, averaging over spatial information, DVMP provides direct sensitivity to the gluon structure of protons

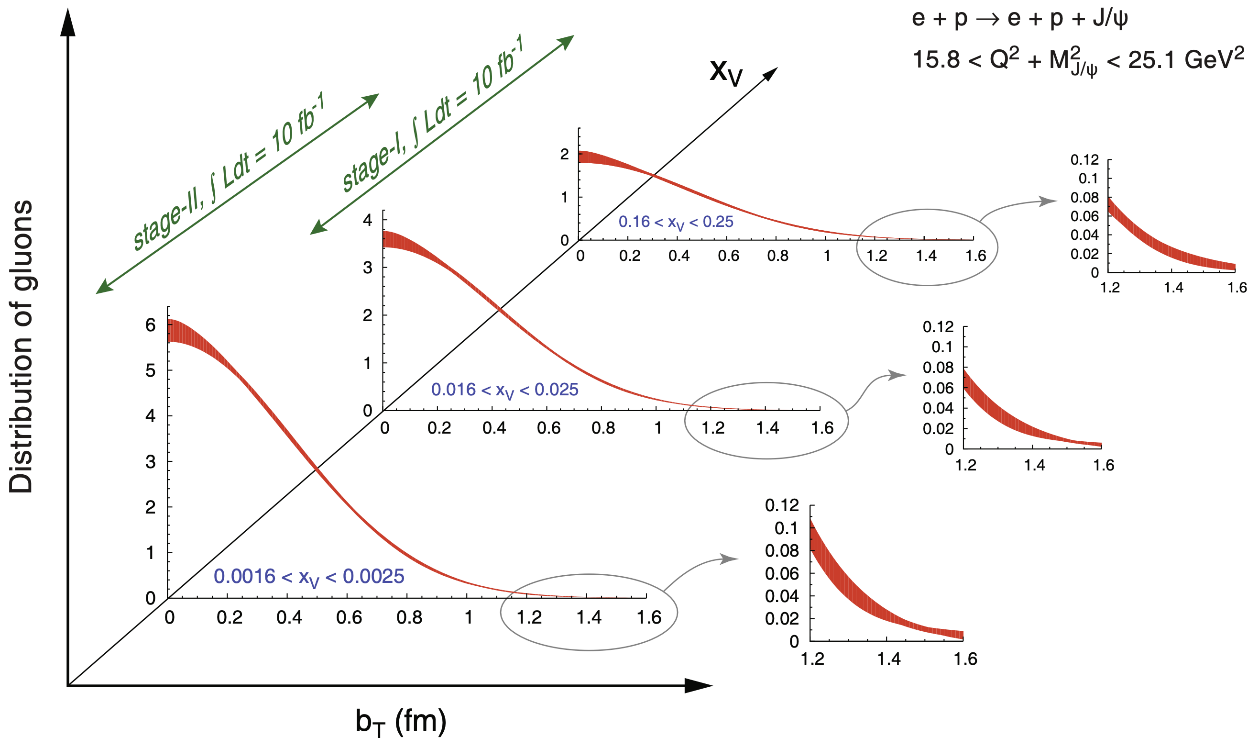

Figure 1: The projected EIC uncertainties for the gluon impact parameter distributions, derived from a Fourier transform of the differential cross-section for J/ψ production as projected in the EIC white paper [1, 2]

as a function of both the gluon momentum fraction x_v (x_v is a modified $Bjorken-x$ variable defined as $x_v = \frac{Q^2+M_V^2}{2P\cdot q}$) and impact parameter b_T from the proton center [1]. By changing x_v, we can explore how gluons are distributed within protons/nucleons at different momentum scales, hence probing the transition from the valence-rich region to the dense region of small-x domain.

The projected precision shown in Figure 1 illustrates the long-term capability of the EIC to perform spatial imaging of gluons through exclusive J/ψ production measurements. The present work is focus in the early science projections, hence it does not yet constitute a full projection study of gluon impact-parameter distributions in the EIC. This study also focuses on following the physics reconstruction framework necessary for such measurements within the ePIC environment, including the reconstruction of exclusive final states, evaluation of detector acceptance and momentum resolution, and studies of the momentum-transfer distribution $d\sigma/dt$ used to extract the spatial distribution of the gluons. To achieve projections comparable to those shown in Figure 1, several additional components are required. These include significantly larger simulated event samples, detailed detector efficiency and systematic uncertainty studies, bin migration and unfolding corrections, and the full Fourier transform analysis needed to obtain impact parameter distributions with quantified uncertainties.

The present analysis therefore represents a step toward the projection in the EIC white paper [1]. These measurements are essential not only for understanding the role of gluons in generating nucleon mass and spin, but also for uncovering signatures of gluon saturation and emergent QCD phenomena in dense systems [2].

2 Simulation Overview and Detector Framework

The ePIC detector is designed to measure particles produced at the interaction point, enabling detailed studies of the internal structure of protons, neutrons, and nuclei. This detector is divided into several regions; the analysis presented concentrates on the acceptance of the Far-forward region with the subsystems' acceptances represented in Table 1.

Detector	**Acceptance**
Zero-Degree Calorimeter (ZDC)	$\theta < 5.5$ mrad $(\sim \eta > 6)$
Roman Pots (RP)	$0.0^{\star} > \theta < 4.7$ mrad $(\sim \eta > 6)$
Off-Momentum Detectors (OMD)	$0.0 > \theta < 5$ mrad $(\sim \eta > 6)$
B0 Detector	$6.0 < \theta < 20$ mrad $(\sim 4.6 < \eta < 5.9)$

Table 1: Summary of Detector Acceptances for the EIC Far-Forward System [6]. This table summarizes the acceptance ranges and specific considerations for various detectors in the EIC far-forward subsystems.

The ePIC far-forward detection system is comprised of multiple components for charged and neutral particle reconstruction. The B0 spectrometer includes an electromagnetic calorimeter for photon reconstruction and tracking detectors for charged particles. Zero-Degree Calorimeters (ZDCs) reconstruct photons and neutrons, while Roman Pots (RPs) and Off-Momentum Detectors (OMDs) track charged particles, with OMDs complementing RPs for lower magnetic rigidity than the main hadron beam (e.g. for spectator protons coming from breakup of light nuclei). The ZDC accepts neutral particles at $\theta < 5.0$ mrad scattering angles. RPs detect forward-scattered protons within approximately $0.2 < \theta < 4.7$ mrad, limited by a minimum transverse momentum of $p_T \geq 0.2$ GeV beam momentum coming from the size of the beam near the Roman pots and minimum safe distance for the detector to operate near the hadron beam. OMDs cover $0.0 < \theta < 5.0$ mrad since it is further from the beam to capture the off-momentum protons, which are bent at larger angles than nominal beam protons in the dipole magnets[2, 6].

2.1 Beam Configuration and Luminosity

To study the feasibility of exclusive J/ψ production at the EIC during the Early Science program, electron–proton collisions at a center-of-mass configuration of 10×130 GeV^2 were simulated. Exclusive J/ψ events, $e + p \rightarrow e' + p' + J/\psi\ (\rightarrow e^+e^-)$, were generated using the lAger event generator [3] with radiative effects enabled. The simulated data correspond to a luminosity of 10 fb^{-1} scaled down to 1 fb^{-1}, obtained from 199,430 generated events processed and "afterburned", which adds the EIC 25mrad crossing angle and also adds realistic beam effects (e.g. angular divergence from focusing of the beam) to provide accurate simulation of overall smearing to momenta from reconstructed particles.[7].

The afterburned events (using preset ip6_ep_130x10 configuration) were subsequently processed through the full ePIC detector simulation using the epic_craterlake detector configuration within the EIC simulation campaign 26.4.1. Detector responses were simulated with ddsim based on the dd4hep geometry framework, and the resulting events were reconstructed using the standard EICrecon software chain. The reconstructed dataset was then used for the final analysis, including event selection, kinematic reconstruction, detector-acceptance studies, and comparisons between Monte Carlo (MC) truth and reconstructed observables.

2.2 Event Generation

lAger, known as the Argonne Generic l/A Event Generator, is a highly adaptable and modular MC simulation framework developed to model a broad range of lepton- and photon-induced reactions on both nucleons and complex nuclei [3]. Designed with versatility in mind, lAger supports simulations across a wide kinematic regime, encompassing both fixed-target configurations, typical of experiments at facilities such as Jefferson Lab, and collider geometries envisioned for the EIC. Significantly, recent developments have focused on enhancing the capabilities of lAger to support DVMP processes, which are of central importance in probing the partonic structure of hadrons and nuclei via GPDs. The generator has been adapted to incorporate models for exclusive vector meson production, with particular attention given to the production of heavy quarkonia such as J/ψ. Studies of J/ψ production have primarily targeted the kinematic regime defined by $1 < Q^2 < 10$ GeV^2, which is especially relevant for probing the transition between non-perturbative and perturbative QCD dynamics. The workflow of the simulation and reconstruction chain used in this analysis is summarized in Figure 2. Events are first generated using lAger (hepmc3), and the resulting kinematics are processed by the Afterburner to incorporate beam crossing angle and beam effects - note that the proton beam is slightly offset at an angle of 25 mrad. The events are then passed to ddsim, which, in combination with the detector geometry provided by dd4hep, produces simulated detector responses stored in a podio file. These simulated data are subsequently reconstructed with EICrecon, yielding both reconstructed podio files and user-friendly ROOT files (eicrecon.root) containing histograms and analysis trees. This framework ensures consistency between generated, simulated, and reconstructed data, providing the foundation for the physics studies presented in this work.

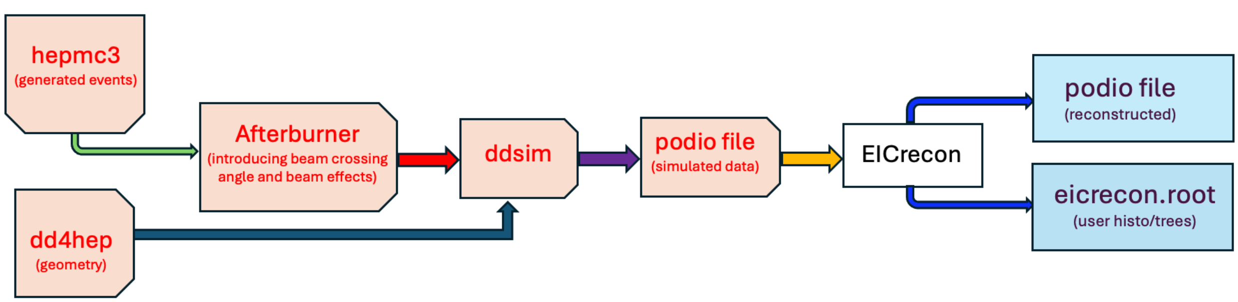

Figure 2: Workflow of the simulation and reconstruction chain [8]. Generated events (HepMC3) from lAger are processed with beam effects using the Afterburner and passed through the detector simulation (ddsim) based on the dd4hep geometry. The simulated data are then reconstructed with EICrecon, producing both podio and ROOT output files for physics analysis.

2.3 Detector Simulation and Generator Level Kinematics

Generator Level Kinematic Ranges:

- $0 < x < 1$,
- $0.1 < y < 0.95$,
- $1 < Q^2 < 10$ GeV^2,

lAger Version used: 3.6.1 – v3.6.x Tweaked VMD formalism and better examples for DVMP at EIC.

Installation instructions and documentation are available in Ref. [3]. We use the 26.4.1 simulation campaign files for the analysis plots.

This project focuses on simulations using the EIC 00 mrads configuration, i.e., events generated without a beam crossing angle. The generator files used as input are in .json format, which allows for detailed customization of event kinematics. Figures 3 and 4 show some of the generator-level kinematics. At the MC level for the final-state particles, Figure 5 shows the MC and reconstructed η coverage in each detector region.

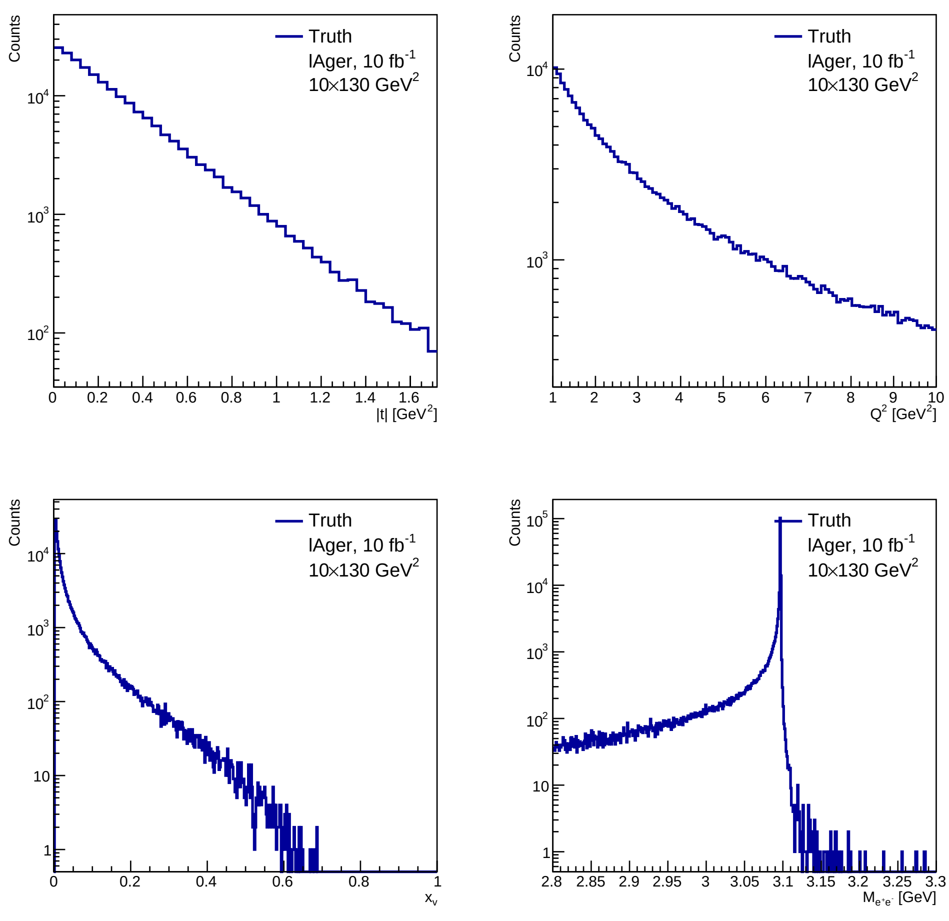

Figure 3: Generator truth-level distributions generated directly with lAger for 10×130 GeV.

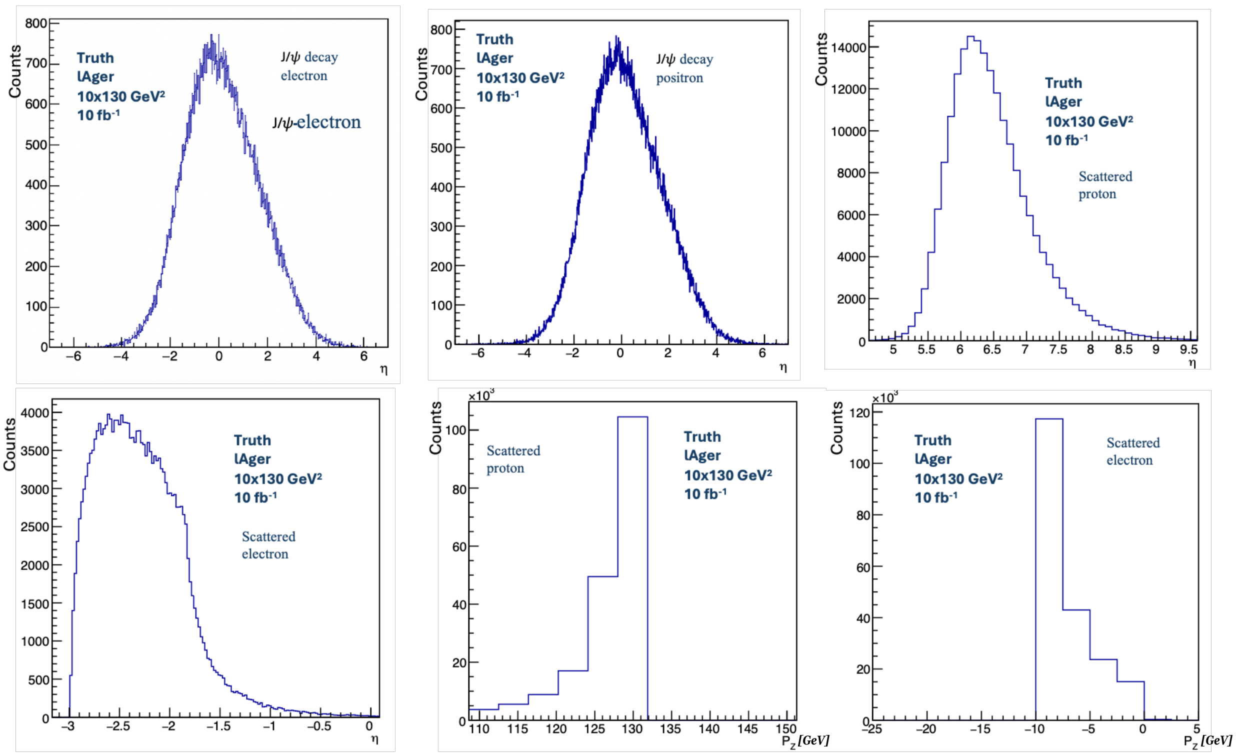

Figure 4: Further generator truth-level distributions generated directly with lAger.

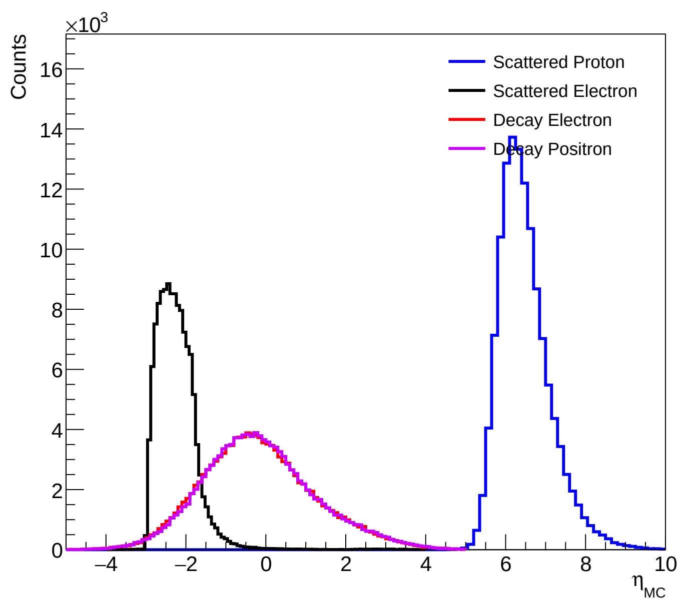

Figure 5: η distributions of MC final-state particles. The coverage of protons (blue), scattered electrons in the backward direction (black), decay electrons in the central detector (red), and positrons (violet) also in the central region, illustrating the detector acceptance across the full η range.

3 Kinematic Reconstruction in Deeply Virtual Meson Production (DVMP)

This section introduces the key physics variables used in this analysis and outlines the baseline kinematic selections applied to the data. Events are required to satisfy $0.1 < y < 0.95$, $1 < Q^2 < 10$ GeV^2, and $0.9 < E/p < 1.2$ (for the scattered electrons), ensuring a kinematic region with reliable detector acceptance and well-defined deep inelastic scattering (DIS) kinematics. Additional exclusivity requirements used to isolate the *ep* exclusive scattering channel are discussed in a subsequent section.

3.1 Definition of Kinematic Variables

An accurate characterization of electron–ion collisions at the EIC necessitates reliable reconstruction of key kinematic variables, including the Bjorken scaling variable x, the negative squared four-momentum transfer Q^2, and the inelasticity y. These observables are fundamental for describing the underlying dynamics of DIS and are critical for achieving the desired precision in the extraction and interpretation of the relevant physics.

- x $(Bjorken - x)$ *or* x represents the fraction of the longitudinal momentum of the nucleon carried by the struck parton (quark or gluon). It provides direct insight into the partonic structure of nucleons and nuclei.
- Q^2 quantifies the resolution of the probe, how deep we are poking into the nucleus: higher Q^2 values correspond to shorter distance scales, allowing the investigation of the internal structure of hadrons with greater detail.
- y (*inelasticity*) indicates the fraction of the electron's energy transferred to the hadronic system in the target rest frame. It is important to determine the energy flow in the event and impacts the sensitivity of different kinematic reconstruction methods.

The next sections explore the above kinematics using four different reconstruction methods. Table 2 presents the explicit mathematical formulations of the variables associated with the different methods under consideration.

3.2 Analysis Code

The analysis code developed for studying gluon distributions through exclusive J/ψ production can be found in Ref. [9].

3.3 Choice of Reconstruction Method

$$e(k) + p(p) \to e'(k') + p'(p') + VM(J/\psi) \tag{1}$$

Deeply-virtual J/ψ production entails reconstructing scattered protons, electrons, and J/ψ decay leptons to compute J/ψ kinematics from their momentum vectors $-t,\ x,\ Q^2$.

Several reconstruction methods are available to study kinematic variables. In this analysis, we considered four reconstruction methods, namely the Electron Method, the Jacquet–Blondel

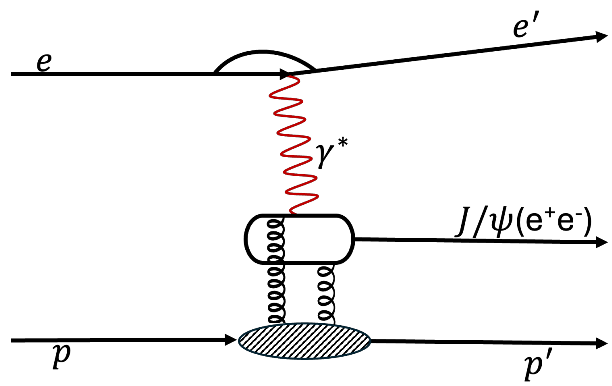

Figure 6: Exclusive process at the EIC illustrating J/ψ production with the e^+e^- decay channel [2]. This topology requires precise measurements of all final-state particles to ensure event exclusivity. Other J/ψ decay channels are also possible but are not shown here.

	x (Bjorken $-\,x$) or x	Q^2	y (inelasticity)
Electron	$\frac{Q^2}{2P\cdot q}$	$-(k-k')^2$	$\frac{P\cdot q}{P\cdot k}$
Jaquet-Blondel (JB)	$\frac{Q^2}{4E_eE_py}$	$\frac{p_{T,h}^2}{1-y}$	$\frac{\Sigma_h}{2E_e}$
Double Angle (DA)	$\frac{Q^2}{4E_eE_py}$	$\frac{4E_e^2}{\tan(\theta_e/2)(\tan(\theta_e/2)+\tan(\gamma/2))}$	$\frac{\tan(\gamma/2)}{\tan(\theta/2)+\tan(\gamma/2)}$
Electron-Sigma ($e\Sigma$)	$\frac{Q_\Sigma^2}{4E_eE_py_h}$	$\frac{p_{T,e}^2}{1-y_\Sigma}$	$\frac{Q_e^2}{4E_eE_px_\Sigma}$

Table 2: Electron vertex kinematic methods comparison. Here E_e and E_p are the energies of the electron and proton beams, respectively [10, 11].

(JB) Method, the Electron-Sigma ($e\Sigma$) Method, and the Double-Angle (DA) Method. Their expressions are summarized and compared in Table 2.

In the following, we compare the kinematic quantities (x), (y), and (Q^2) reconstructed from the corresponding ROOT-file branches InclusiveKinematicsJB, InclusiveKinematicsESigma, InclusiveKinematicsDA, and InclusiveKinematicsElectron. Here, "inclusive" denotes the kinematic reconstruction based solely on the scattered electron from the TTree branch, i.e., the process $e + p \to e' + X$, in which all other final-state particles (X) are not explicitly reconstructed. This comparison is motivated by the fact that the inclusive reconstruction yields an appropriate and unbiased determination of the underlying kinematic variables.

Other variable definitions,

- the transverse momentum of the Hadronic Final State,

$$p_{T,h}^2 = \left(\sum p_x\right)^2 + \left(\sum p_y\right)^2 \tag{2}$$

- the energy imbalance of the Hadronic Final State,

$$\Sigma_h = \sum E - \sum p_z \tag{3}$$

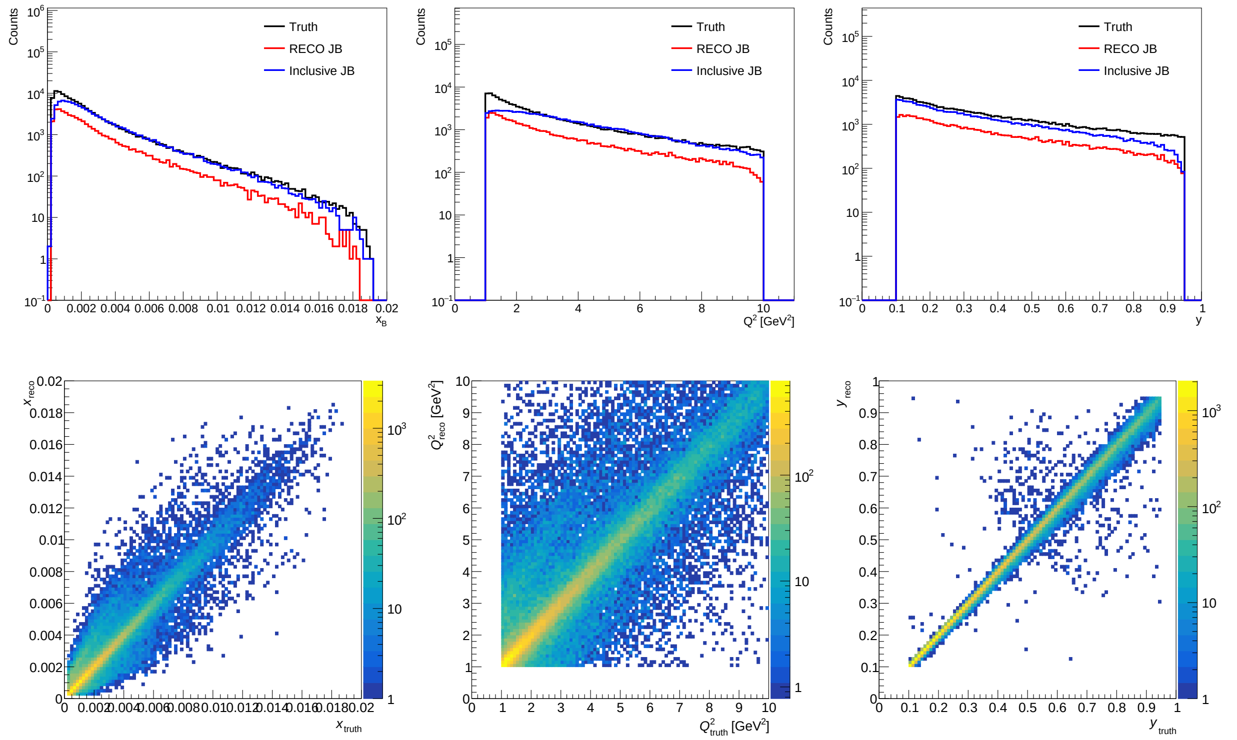

Figure 7: Electron-vertex kinematic variables reconstructed with the JB method. From left to right: x, Q^2, and y. **Top row:** MC truth, RECO, and Inclusive TTree branch distributions from EICrecon. **Bottom row:** MC-truth versus RECO correlations.

- the hadronic angle γ,

$$\tan(\gamma/2) = \Sigma_h/p_{T,h} \tag{4}$$

- the electron energy imbalance of the scattered electron,

$$\Sigma_e = E - p_z \tag{5}$$

- total energy imbalance,

$$\Sigma_{tot} = \Sigma_e + \Sigma_h \tag{6}$$

- using Electron and Σ we can derive the following needed for $e\Sigma$ method,

$$y_e = 1 - \frac{\Sigma_e}{2E_e};\ Q_e^2 = \frac{p_{T,e}^2}{1 - y_e};\ y_\Sigma = \frac{\Sigma_h}{\Sigma_{tot}} \tag{7}$$

These variables - Q^2, x, and y - form a good description of the DIS kinematics, allowing a detailed investigation of quark and gluon distributions within the target beam at varying scales. Figure 6 illustrates the reaction from which the kinematic expressions are derived. Figures 7, 8, 9, and 10 present the reconstructed kinematic distributions obtained using the different reconstruction techniques. Each method exhibits distinct sensitivities to specific detector subsystems and to radiative effects. The resolution comparison of the different

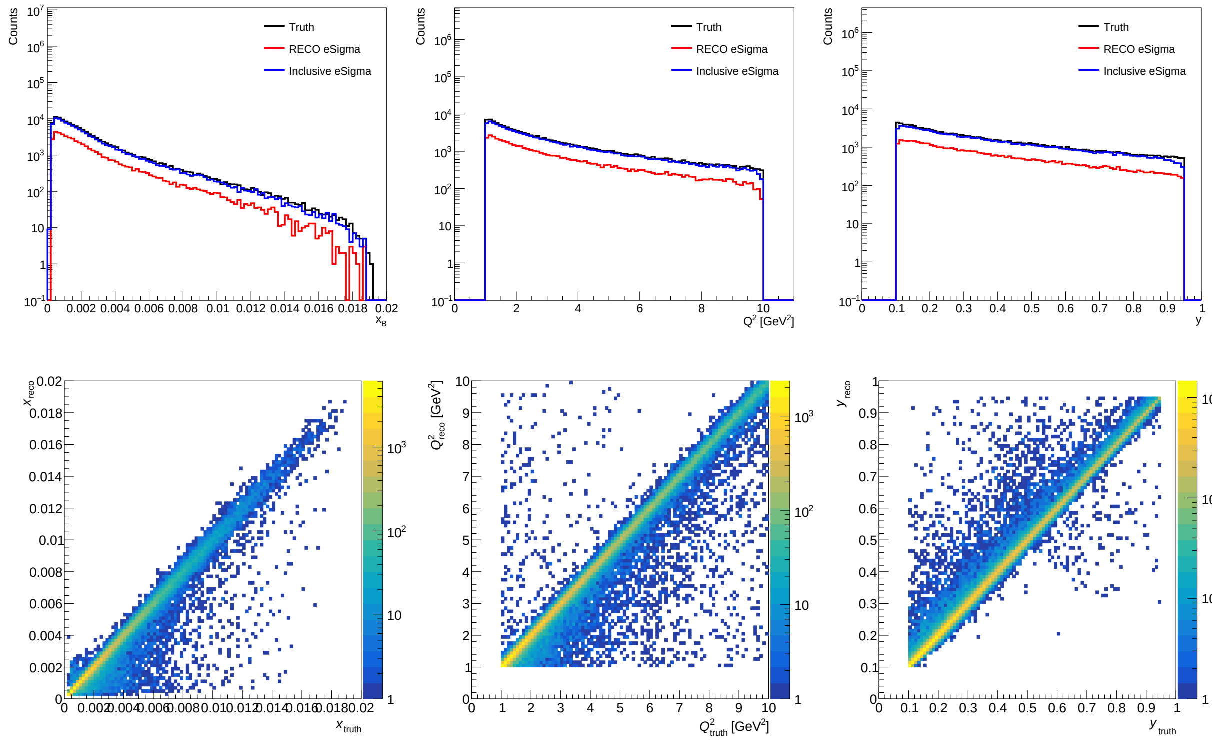

Figure 8: Electron-vertex kinematic variables reconstructed with the $e\Sigma$ method. From left to right: x, Q^2, and y. **Top row:** MC truth, RECO, and Inclusive TTree branch distributions from EICrecon. **Bottom row:** MC-truth versus RECO correlations.

methods is shown in Figure 11, and we can observe that the DA method has the best resolution.

The JB method uses only the hadronic final-state information and is particularly useful when the proton is detected. However, its performance is limited by the resolution of the hadronic system. The resolution plots show a broader spread for x and Q^2, while the reconstruction of y is improved but still exhibits broader tails than the other methods.

The $e\Sigma$ method combines information from both the scattered electron and the hadronic final state, leading to improved reconstruction across all kinematic variables. Its resolution is generally better than the JB method but slightly broader than the DA method.

The Electron method relies only on the scattered electron, whose energy and momentum are typically well measured. The resolution plots show good reconstruction performance, with relatively narrow distributions centered near zero for all three kinematic variables.

The DA method combines information from both the scattered electron and the hadronic system. As a result, it provides the narrowest and most stable resolution for x, Q^2, and y.

In this analysis, the choice of reconstruction method is guided by the resolution performance, as in Figure 11 and the availability of reconstructed final-state particles. The DA method is preferred when both scattered-electron and hadronic information are available, while the Electron method is used in cases where the hadronic final state is only partially reconstructed. Even when the recoil proton is not directly detected, information about its four-momentum can still be inferred from energy–momentum conservation using the recon-

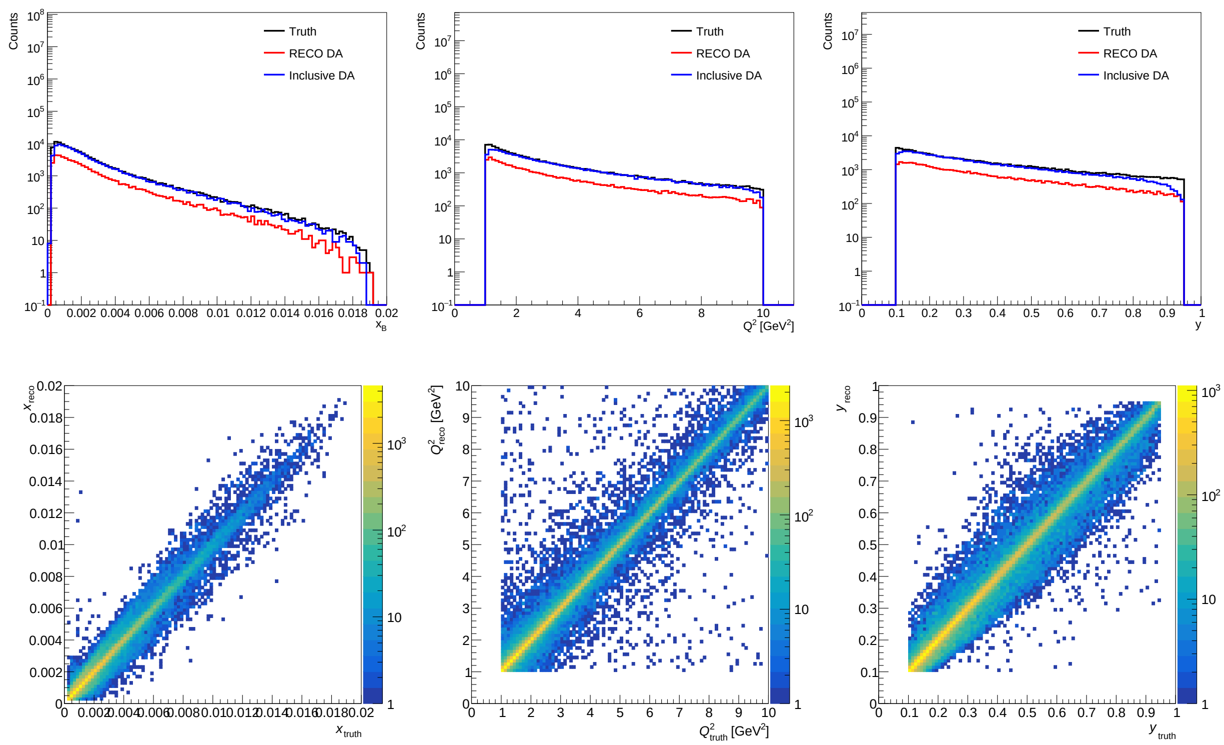

Figure 9: Kinematic variables reconstructed with the DA method. From left to right: x, Q^2, and y. **Top row:** MC truth, RECO, and Inclusive TTree branch distributions from EICrecon. **Bottom row:** MC-truth versus RECO correlations.

structed scattered electron and J/ψ channel. Such missing-momentum techniques provide an additional constraint on the event kinematics and are further explored in the corrected recoil reconstruction method discussed in Section 5.3.

4 Event Selection and Exclusivity

For exclusive DVMP analysis, strict event exclusivity is imposed, requiring that all final-state particles in the reaction, as specified in Eq. 1 and Figure 6, be fully reconstructed. The scattered electron and the J/ψ decay leptons are detected in the central detector subsystems, whereas the recoil proton is measured in the far-forward instrumentation, comprising the B0 spectrometer and the RP detectors. Experimentally, exclusivity is validated through the simultaneous reconstruction of all final-state particles, which enables strong suppression of backgrounds arising from proton dissociation and other non-exclusive reactions.

Precise identification and selection of the scattered electron and the recoil proton are crucial for the robust reconstruction of the principal kinematic variables, namely x, Q^2, and y (as illustrated in Figures 7, 8, 9, and 10), as well as the squared four-momentum transfer t (cf. Figure 17). Stringent and optimized selection criteria enhance the resolution of these observables, mitigate systematic uncertainties, and ensure mutual consistency among the different kinematic reconstruction methods.

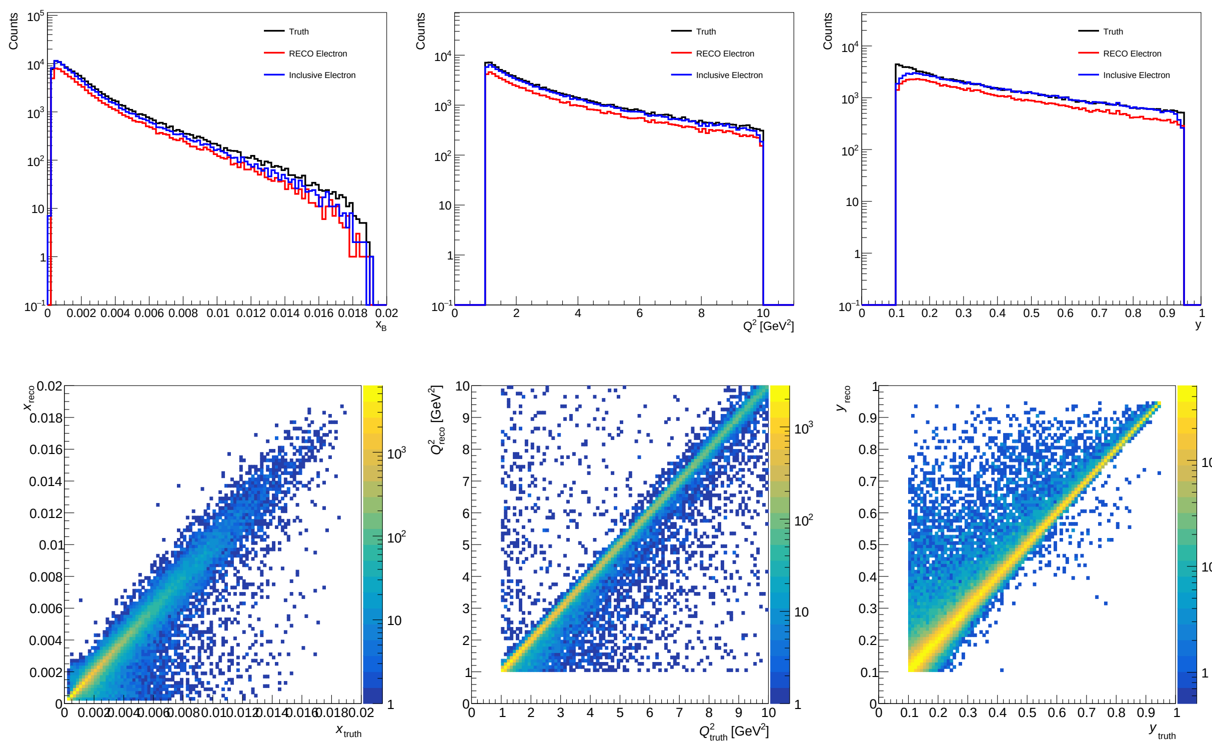

Figure 10: Electron-vertex kinematic variables reconstructed with the Electron Method. From left to right: x, Q^2, and y. **Top row:** MC truth, RECO, and Inclusive TTree branch distributions from EICrecon. **Bottom row:** MC-truth versus RECO correlations.

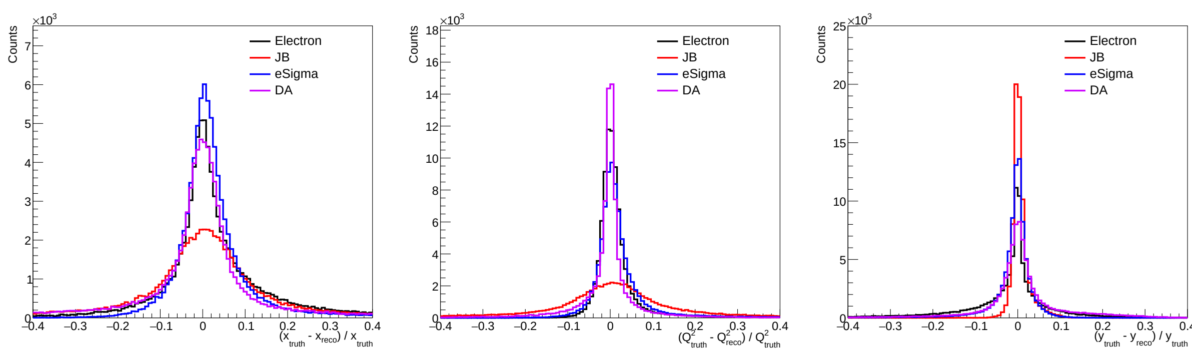

Figure 11: Comparison of the relative reconstruction residual, $\Delta^{\mathrm{rel}} = \frac{\mathrm{truth}-\mathrm{RECO}}{\mathrm{truth}}$, for x (**Left**), Q^2 (**Middle**), and y (**Right**) using the Electron, JB, $e\Sigma$, and DA kinematic reconstruction methods. Distributions that are narrower and more strongly centered around zero correspond to improved reconstruction resolution and smaller reconstruction bias. The DA method provides the best overall performance, exhibiting the narrowest and most symmetric distributions for all three kinematic variables. The Electron and $e\Sigma$ methods also achieve good reconstruction accuracy, with only slightly broader residual distributions. In contrast, the JB method shows substantially broader distributions, particularly for x and Q^2, reflecting its greater sensitivity to detector effects and hadronic energy resolution.

The following subsections describe the selection criteria applied to each category of final-state particles and the resulting acceptance characteristics.

4.1 Electron Identification

To reconstruct the scattered electron, decay electron, and the decay positron, this analysis uses the ReconstructedTruthSeededChargedParticles branch, from which the reconstructed particle kinematic properties are obtained. These include the energy E, mass m, charge q, and momentum components (p_x, p_y, p_z), which are used to construct the electron four-momentum and apply the electron identification.

Among the three final-state leptons, two distinct electrons are present, corresponding to the decay electron and the scattered electron. It is crucial to differentiate these two particles. To this end, two electron candidates, denoted as electron-vector-1 and electron-vector-2, are analyzed. For each event, the electron candidates are tested together with the candidate positron to identify the $(J/\psi \to e^+e^-)$ decay pair. Their invariant masses and momentum properties are used to distinguish the decay electron from the scattered electron. The reconstructed particle is selected as the scattered electron if it has associated tracks and clusters and satisfies the following criteria:

- Negatively charged
- An isolation requirement is applied to the electron candidate. The leading cluster is accepted if its energy, E_0, is greater than the total energy of all other clusters within a cone of $\Delta R < 0.4$ [1].
- If either invariant mass lies within the window $2.9 < M_{J/\psi} < 3.2$ GeV, the electron with the lower absolute longitudinal momentum ($|P_z|$) is assigned as the **decay electron** originating from the J/ψ.
- If the invariant mass falls outside this window, the electron with the higher $|P_z|$ is identified as the **scattered electron**.
- The scattered electron is further required to satisfy an energy–momentum consistency cut (note that this condition is satisfied for all energy and momentum of reconstructed leptons), as shown on the left panel of Figure 13:

$$0.9 < E/p < 1.2$$

Events for which the selection process is not successful, due to the failure to satisfy the requirements, are rejected. For consistency of results, any condition applicable to the MC must be equally applicable to the RECO information. A comparison of this identification technique with the electron from the backward endcap electromagnetic calorimeter, as shown in the middle panel of Figure 13, shows our reconstruction is efficient. Figure 13 presents a comparison of the energy resolution obtained from tracks and clusters in the kinematic range $1 < Q^2 < 10$.

[1] $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$, where $\Delta\eta$ is the difference in pseudorapidity and $\Delta\phi$ is the difference in azimuthal angle between the electron candidate and a neighboring cluster.

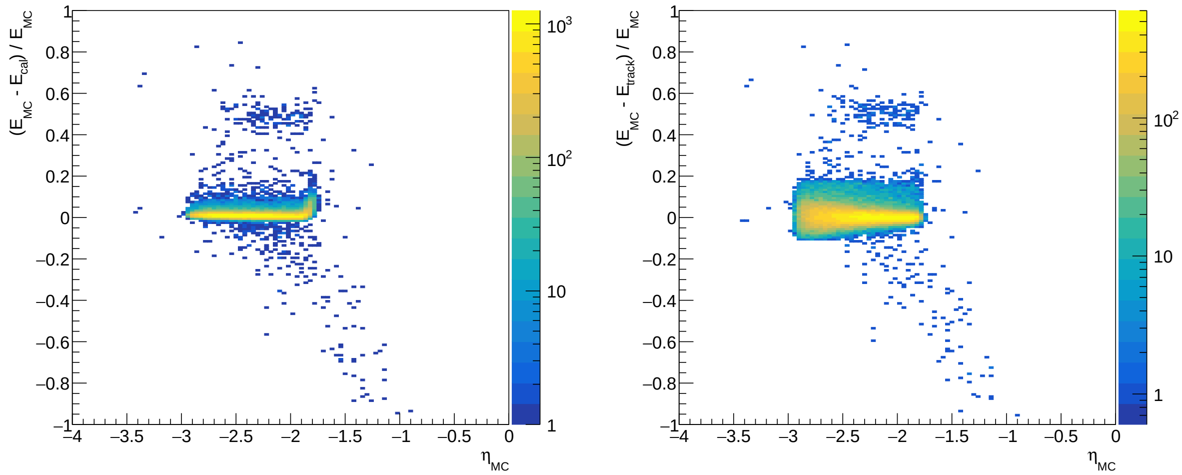

Figure 12: Quality of the scattering electron energy reconstruction as a function of true η, **Left:** Energy resolution of cluster. **Right:** Energy resolution of tracking.

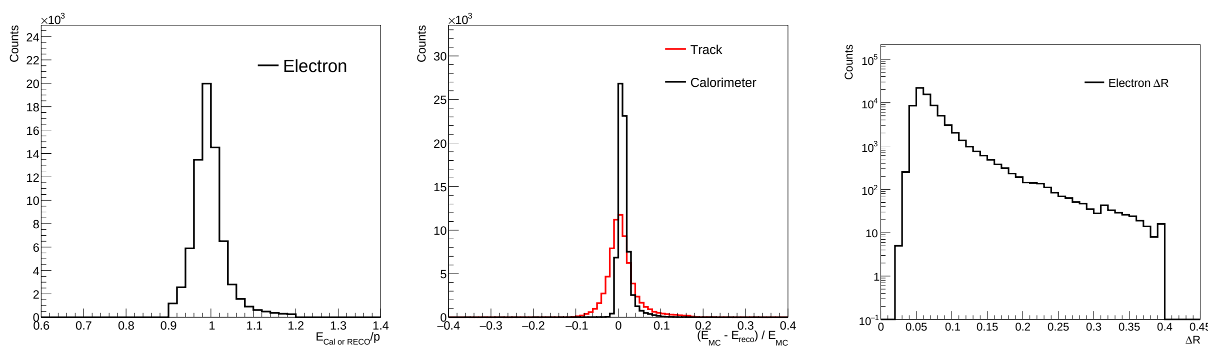

Figure 13: **Left:** Distribution of cluster electromagnetic calorimeter energy-to-reconstructed (track) momentum ratio for e', E/p, exhibiting a pronounced peak around unity as expected for well-identified electrons. **Middle:** Relative energy resolution, $\Delta_{\text{Energy}} = (\text{Truth} - \text{RECO})/\text{Truth}$, comparing track-based (red) and calorimeter-based (black) measurements. The narrow, centered peaks indicate good energy reconstruction performance, with the calorimeter showing improved resolution. **Right:** Distribution of the electron isolation variable, ΔR, used in the electron selection. Electron candidates are required to satisfy an isolation cone of $\Delta R < 0.4$, ensuring that the candidate is well separated from nearby calorimeter activity.

In events where an energy cluster was reconstructed in the backward electromagnetic calorimeter, the cluster energy was used to determine the electron four-momentum, while the spatial coordinates were derived from the tracking system. To achieve an optimized reconstruction of the scattered electron, the energy input to the four-momentum calculation was chosen on an event-by-event basis from either the calorimeter or the tracking system, according to the performance comparison presented in Figure 12.

An additional selection based on the longitudinal energy balance, $E - p_z$, is applied to ensure consistency with well-reconstructed exclusive DIS kinematics. In the absence of significant detector losses or radiative effects, energy–momentum conservation requires that the summed quantity $\sum(E - p_z)$ over all final-state particles peaks near $2E_e$, as illustrated in the left panel of Figure 14, where E_e is the incoming electron beam energy. Deviations from this expectation typically indicate non-exclusive backgrounds, poorly reconstructed particles, detector acceptance losses, or significant QED radiation. The $E - p_z$ requirement therefore improves the purity of the selected exclusive J/ψ sample and enhances the stability of the reconstructed kinematic variables.

In addition, Figure 14 right panel shows the reconstructed invariant mass, which is in the range $2.9 < M_{J/\psi} < 3.2$ GeV.

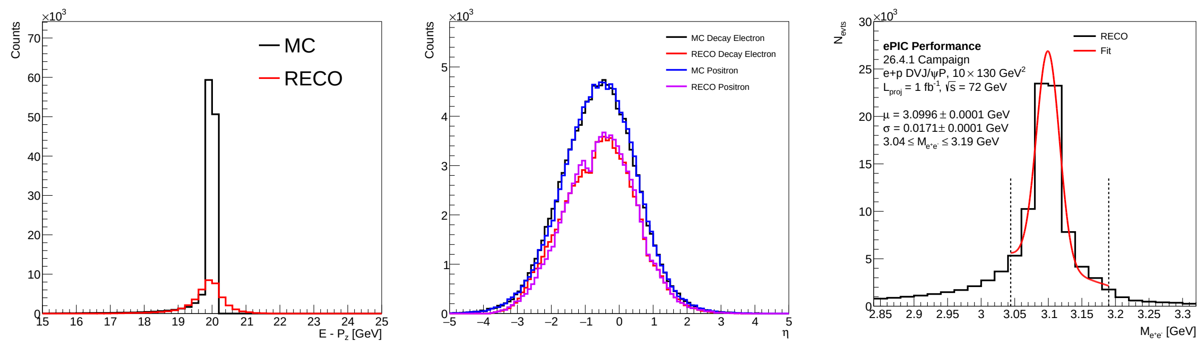

Figure 14: **Left**: Sum of $E - p_z$ over all final-state particles in the event. **Middle**: η distributions of the decay leptons from $J/\psi \to e^+e^-$ at generator (MC) and RECO levels, shown separately for electrons and positrons. The reconstructed distributions closely follow the MC shapes, with reduced yields due to detector acceptance and reconstruction effects. **Right**: Reconstructed invariant mass distribution of the e^+e^- pair. The red curve shows a Gaussian fit to the J/ψ signal peak, yielding a mean mass consistent with the PDG value and a mass resolution of $\sigma \approx 17.1 \pm 0.1$ MeV. The dashed vertical lines indicate the invariant mass window used for J/ψ event selection.

The selection criteria defined at the MC level were implemented at the reconstruction level, with the additional requirement of a stringent multiplicity cut: events were required to contain exactly three reconstructed charged-particle tracks, and any event not satisfying this condition was rejected.

4.2 Proton Selection

The squared four-momentum transfer t is a key observable in exclusive J/ψ production and is defined as

$$t = (p - p')^2, \tag{8}$$

where p and p' are the four-momenta of the incoming and scattered proton, respectively. In this analysis, t is evaluated at both the MC truth level and the RECO level. These are treated separately, as they rely on different inputs and selection criteria.

4.2.1 MC-Level Proton Selection

At the MC level, the event kinematics are reconstructed using the MCParticlesHeadOnFrameNoBeamFX branch, which stores the generated final-state particles in the head-on collision frame prior to the inclusion of beam effects. For the particle, the reconstructed quantities include the momentum components (p_x, p_y, p_z), E, m, q, and the generator status flag (generatorStatus). These quantities are used to identify the incoming and scattered particles participating in the exclusive reaction and to construct the corresponding four-momenta for the evaluation of the event kinematics and momentum-transfer variable t. The generator status information is further employed to distinguish beam particles from stable final-state particles, ensuring a consistent MC-level definition of the exclusive topology.

The incoming proton beam is identified using particles with status code 4 and PDG ID 2212, while the scattered proton is selected from stable final-state particles (status code 1) with PDG ID 2212; Figure 15 shows the typical MC proton η plot in black. The momentum transfer is calculated using the following selection criteria:

- Incoming proton: status = 4, PDG = 2212, charge = +1
- Scattered proton: status = 1, PDG = 2212, charge = +1

4.2.2 Recoil Proton Detection in the Roman Pots

When the proton is detected in the RP detectors, reconstruction is performed using (p_x, p_y, p_z), E, m, and q branches of the ForwardRomanPotRecParticles, which gives a fully reconstructed momentum vector from the detector hits. The pseudorapidity range of RP is from $\eta > 6$, as shown in Figure 15, where the angle coverage is $0 < \theta < 4.7$ mrad. The event selection process here focuses on identifying valid proton candidates in the Roman Pot (RP) detector and computing the four-momentum transfer t using the equation $t = (p - p')^2$.

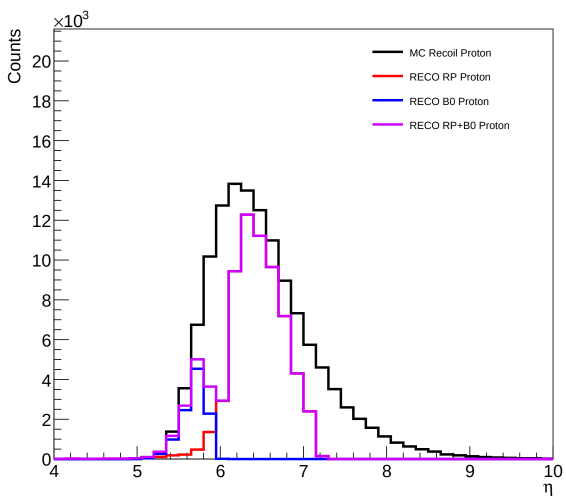

Figure 15: η distribution of the recoil proton comparing MC truth (black) with RECO tracks in the RP (red), B0 (blue), and RP+B0 (violet).

- The code iterates over all events in the dataset (the event loop), and for each event, all reconstructed particles are iterated over (the particle loop).

- RP tracks are obtained from the 4 branch, where the proton momentum is reconstructed from the detector hit information.

- At the reconstructed level, the RP reconstruction algorithm already assumes the detected particle corresponds to a forward-scattered proton. Therefore, the proton candidate is selected from the reconstructed RP track with positive charge.

- A transverse momentum cut of $p_T > 0.2$ GeV is applied to suppress poorly reconstructed tracks.

- Since the proton beam collides with the electron beam with a crossing angle of 25 mrad along the y-axis, this effect is corrected by applying a counter-rotation to the reconstructed proton momentum.

4.2.3 Recoil Proton Detection in the B0 Subsystem

For events in which the recoil proton is detected in the B0 spectrometer, the relevant branch is ReconstructedTruthSeededChargedParticles. This algorithm reconstructs tracks from B0 detector hits and stores the output in the TruthSeededReconstructedChargedParticles collection. Currently, B0 reconstruction relies on truth seeding because the default tracking algorithm used for the ePIC detector does not yet perform reliably in the B0 region. The track reconstruction in the B0 tracker is performed using the ACTS framework [12].

- Loop over all events in the dataset.

- For each event, loop over the reconstructed charged particles in the B0 track collection.

- Apply a pseudorapidity selection of $\eta > 4$, corresponding approximately to the angular acceptance range $6 < \theta < 20$ mrad.

- Require a positive charge (+1) to select proton candidates.

- Correct for the beam crossing angle of 25 mrad along the y-axis by applying a counter-rotation to the reconstructed proton momentum.

4.3 Missing Mass Requirement

To further enhance the purity of the exclusive sample, an additional selection is applied based on the missing mass of the event. This cut is specifically applied when constructing the t-distribution, ensuring that only events consistent with the exclusive reaction Eq. 1 are included. By restricting the missing mass to $|M_X| < 2$ as shown in Figure 16, where MC gives the truth information, background contributions from non-exclusive processes are suppressed, improving the fidelity of the reconstructed t-distribution.

For the reaction, the method implements energy conservation and momentum conservation laws [13], a check whether the detected final-state particles account for all the momentum in the event. With this, we will ideally expect that if event is exclusive (no other particles are produced), the missing four-momentum should vanish.

$$M_X = P_e + P_p - (P_{e\prime} + P_{p\prime} + P_{J/\psi}) \tag{9}$$

M_X is effective for removing inelastic background.

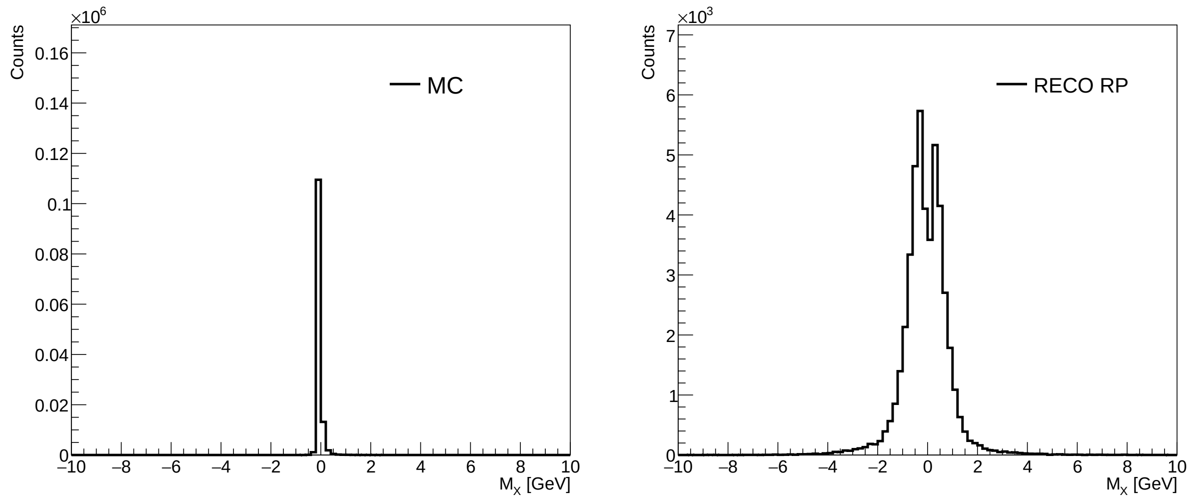

Figure 16: Typical M_X using 26.4.1 10×130 lAger sample; **Left** - MC, **Right** - reconstructed RP.

4.4 Event Reconstruction Exclusivity

The exclusivity selection in reconstructed events is implemented through a workflow designed to identify all particles in the final state, requiring that e', p', and $J/\psi(e^+e^-)$ be found to estimate the distribution of $-t$. The reconstruction procedure is summarized as follows:

1. Identify electron candidates satisfying the selection criteria discussed in Section 4.1.
2. Identify recoil proton candidates satisfying the selection requirements described in Section 4.2 using either the RP or the B0 detector systems. In events where proton candidates are reconstructed in both the RP and B0 detectors, the RP reconstruction is prioritized due to its superior reconstruction performance in the low-$|t|$ region.
3. Events in which no recoil proton candidate is reconstructed in either the RP or B0 detectors are excluded from the direct proton-recoil reconstruction method and are considered only in approaches based on inferred missing momentum.
4. Use the selected events to reconstruct the relevant kinematic observables, particularly the momentum transfer $|t|$, and populate the corresponding analysis distributions.

This procedure ensures that the reconstructed event topology remains consistent with the expected exclusive DVMP final state while maximizing the acceptance coverage provided by the complementary RP and B0 far-forward detector systems.

5 Reconstruction of Momentum Transfer

5.1 Physics of Momentum Transfer

The quantity $-t$ involves essential information about the spatial structure of the proton. Since $-t$ is related to the impact parameter via a mathematical relation (a Fourier transform), measuring its distribution provides direct access to the transverse spatial distribution of partons within the nucleon. Studying the t-distribution allows us to investigate correlations between the transverse positions of partons and their longitudinal momentum fractions, enabling a tomographic view of the nucleon. An accurate reconstruction of $-t$, particularly for small-angle (events at proximity to the beamlines), far-forward scattered protons, is therefore crucial for probing the gluon spatial structure and testing QCD-based models of exclusive processes.

This analysis uses complementary approaches to reconstruct the (t)-distribution, addressing the limitations of the RP and B0 subsystems as well as regions of reduced detection efficiency. The motivation for using these two methods, together with their respective domains of applicability, is discussed in the following sections.

5.2 Method 1: Proton Recoil Method

In exclusive processes, the squared four-momentum transfer to the proton is given by,

$$t = (p - p')^2$$

where p is the initial proton four-momentum and p' is the four-momentum of the recoil proton after the interaction. In the proton recoil method, p' is directly measured using tracking detectors to detect the scattered proton. The initial proton four-momentum p is determined from the nominal beam energy and beam direction, including the known beam-crossing geometry and beam-smearing conditions implemented in the simulation. Figure 17 shows events reconstruction for both the RP and B0 detectors using this method.

	$\lvert t\rvert$ range (GeV^2)	θ range (in mrad)
RP	[0, 0.3733]	[0, 4.7]
B0	[0.6084, 6.7600]	[6.0, 20.0]

Table 3: Kinematic coverage of the far-forward proton detectors in terms of four-momentum transfer $|t|$ and scattering angle θ. The RP detectors measure protons at very small scattering angles ($0 < \theta < 4.7$ mrad), corresponding to the low-$|t|$ region ($|t| < 0.37$ GeV^2). The B0 spectrometer covers larger angles ($6 < \theta < 20$ mrad), corresponding to higher $|t|$ values ($0.61 < |t| < 6.76$ GeV^2). The region between these ranges represents a geometric acceptance gap between the two detectors.

The RP and B0 detectors provide complementary coverage of the recoil-proton phase space. The RP system is optimized for the very small-angle scattering region, corresponding to low four-momentum transfer $|t|$, whereas the B0 detector extends the kinematic acceptance to larger scattering angles and higher values of $|t|$. Owing to the geometric separation between the two detector systems, the gap in $|t|$ coverage between RP and B0 is smeared out here, as shown in Figure 17, where Table 3 shows the $|t|$ range.

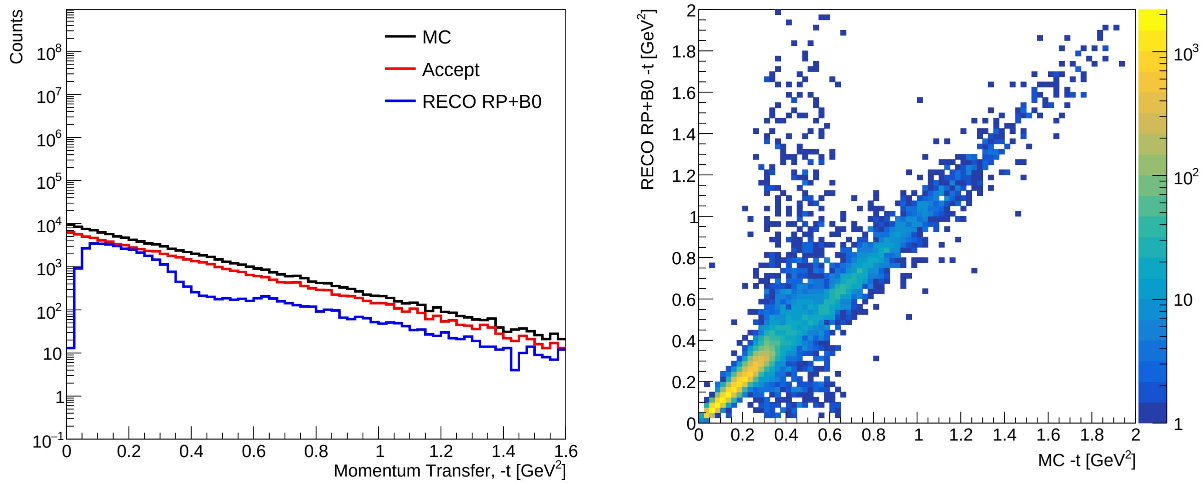

Figure 17: **Left**: Momentum transfer t-distribution, comparing MC truth (black), acceptance (red), and reconstructed events (blue). **Right**: Correlation between reconstructed and true momentum transfer, t_{RECO} versus t_{MC}, demonstrating the resolution and kinematic reach of the recoil proton reconstruction.

5.3 Method 2: Corrected Method through Invariant Mass

In the region between the RP and B0 acceptances, the recoil proton (p') cannot be directly detected. A similar limitation occurs in regions where the detection efficiency falls below 50%. In these cases, an alternative method can be used to determine (t).

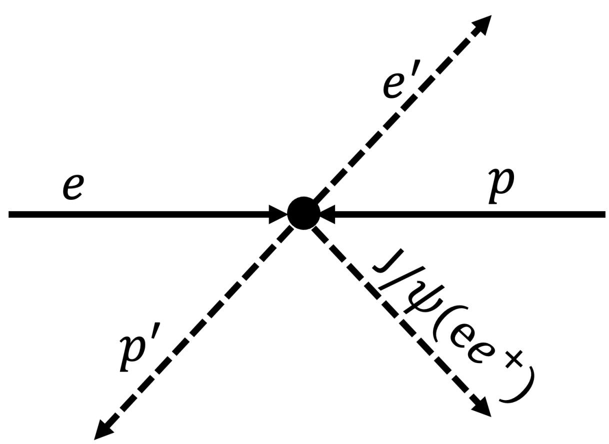

Figure 18: Schematic diagram of the reaction $ep \to e'p'J/\psi$. Reaction products are shown with dashed lines.

We have

$$t = (P_p - P_{p'})^2 \tag{10}$$

where the Figure 18 can be written as

$$e + p \to e' + p' + J/\psi(e^+e^-) \tag{11}$$

deriving the scattered p', we have that;

$$P_{p'} = P_e + P_p - P_{e'} - P_{J/\psi} \tag{12}$$

$$p_p^+ = E_{p'} + p_{z,p'} \tag{13}$$

$$p_{T,p}^2 = p_{T,p'}^2 \tag{14}$$

$$p_p^- = \frac{M_p^2 + p_{T,p}^2}{p_p^+} \tag{15}$$

We then correct the proton's longitudinal and energy components by setting:

$$p_z^{\text{corr}} = \frac{p_p^+ - p_p^-}{2}, \quad E^{\text{corr}} = \frac{p_p^+ + p_p^-}{2} \tag{16}$$

while keeping the same transverse momentum components p_x, p_y.

Finally, we calculate the Mandelstam variable t using the corrected recoil proton, as shown in Figure 19:

$$t_{\text{Method 2}} = -\left[\left(P_p - P_{p'}^{corr}\right)^2\right] \tag{17}$$

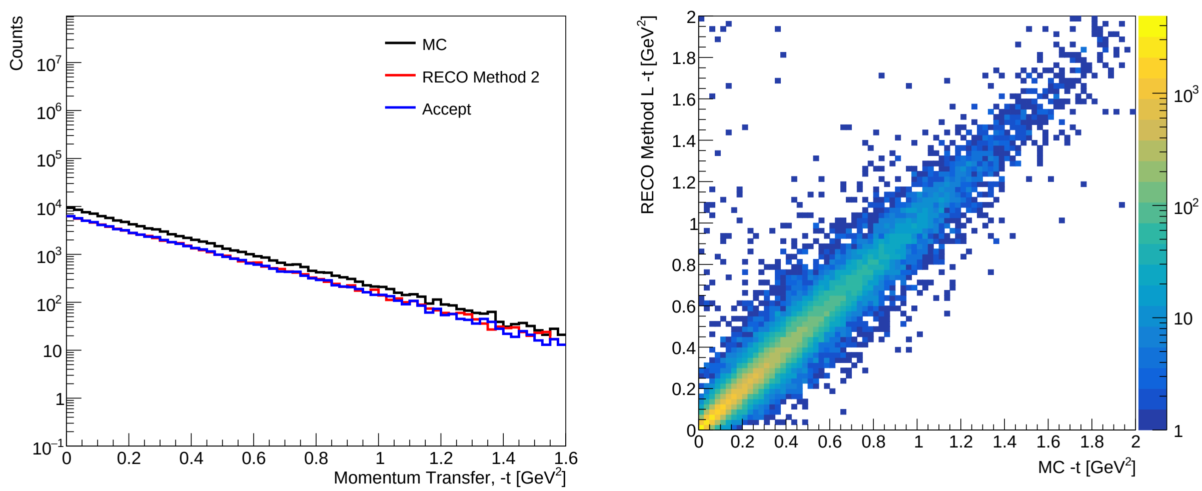

Figure 19: **Left:** Momentum transfer ($-t$) distributions obtained using the corrected proton recoil method, $t_{\text{Method 2}}$, shown for MC truth (black), acceptance-level (blue), and reconstructed events (red). **Right:** Correlation between reconstructed and MC truth t, demonstrating the resolution and linearity of the method. The strong diagonal trend indicates good agreement, validating the use of the corrected recoil technique to bridge the gap between RP and B0 t measurements.

5.4 Combined t-Distribution Using RP, B0, and Method 2

For events reconstructed with the RP and B0 detectors, the four-momentum transfer t is determined using **Method 1**, as shown in Figure 17. To reduce the discontinuity between the RP and B0 measurements in the reconstructed $-t$ distribution, Eq. 17 is used as **Method 2**. This method reconstructs the recoil proton from the corrected missing four-momentum, allowing events to be recovered in regions where direct proton detection is limited by detector coverage and low reconstruction efficiency.

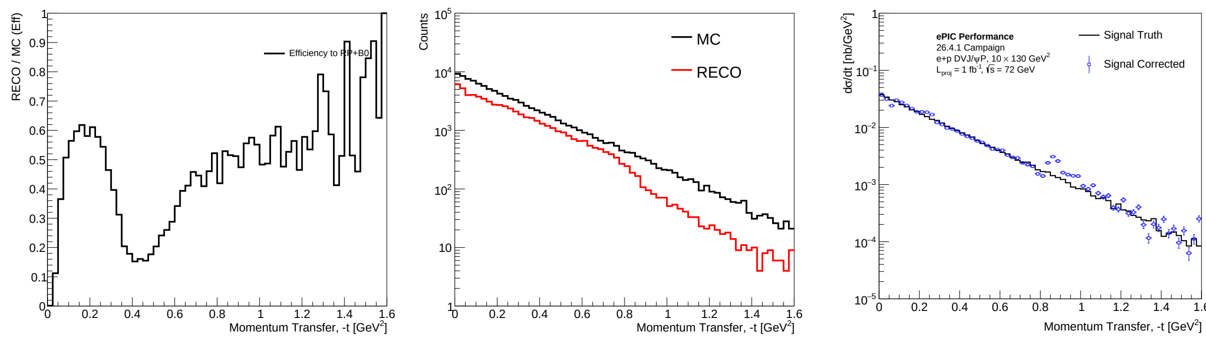

Figure 20: **Left:** Acceptance and reconstruction efficiency versus $-t$ for the combined RP and B0 reconstruction, computed as the ratio of RECO to MC truth events in each $-t$ bin. **Middle:** MC and RECO $-t$ distributions for the combined reconstruction, where **Method 2** covers the low- and intermediate-t regions and RP/B0 reconstruct the recoil proton within their respective acceptances. **Right:** MC truth and acceptance-corrected signal differential cross sections, $d\sigma/dt$, as a function of $-t$.

The reconstructed $-t$ distribution obtained using **Method 1** alone exhibits reduced efficiency at very low $-t$ and in the transition region between the RP and B0 acceptances. To improve the reconstruction in these regions, **Method 2** is incorporated, providing additional acceptance where the far-forward detectors have limited coverage. Figure 19 demonstrates that $t_{\text{Method 2}}$ effectively bridges the acceptance gap between the RP and B0 detectors, particularly in the kinematic region where both detectors exhibit low reconstruction efficiency.

The reconstruction efficiency is shown in the left panel of Figure 20 and is defined as

$$\epsilon(t) = \frac{N_{\text{RECO}}(t)}{N_{\text{MC}}(t)}, \tag{18}$$

where N_{RECO} denotes the number of reconstructed events obtained using the combined RP and B0 reconstruction, while N_{MC} represents the corresponding MC truth distribution after applying the same selection process.

Table 4 summarizes the reconstruction method adopted in each $-t$ region. **Method 2** is used in the low-t region, where direct proton reconstruction is limited, and again in the intermediate transition region to recover events that are not efficiently reconstructed by the far-forward detectors. The RP detector is used where its reconstruction efficiency exceeds approximately 50%, while the B0 detector reconstructs recoil protons at larger momentum transfers. This hybrid approach exploits the strengths of each reconstruction technique, providing continuous coverage and improved statistics across the full accessible $-t$ range.

Table 4: Reconstruction method used in different momentum transfer ($-t$) regions.

$-t$ **Range** [GeV^2]	**Reconstruction Method**
$0.000 \leq -t < 0.078$	**Method 2**
$0.078 \leq -t < 0.270$	RP (**Method 1**)
$0.270 \leq -t < 0.830$	**Method 2**
$0.830 \leq -t < 2.000$	B0 (**Method 1**)

The middle and right panels of Figure 20 show the reconstructed $-t$ distribution and the extracted differential cross section after combining **Method 2**, RP (**Method 1**), and B0 (**Method 1**). Each reconstruction method is applied only within its assigned $-t$ interval, as summarized in Table 4, without overlap or fallback between methods. Consequently, small discontinuities are observed at the boundaries where the reconstruction method changes, as shown in the right panel of Figure 20. The features observed in the joining bins are expected and reflect the different detector acceptances, smearing effects, reconstruction efficiencies, and bin migrations associated with **Method 1** and **Method 2**.

6 Acceptance Corrections and Cross Section Extraction

6.1 Acceptance Correction Factor

Using the combined spectrum, the effects of detector acceptance are taken into account by applying an acceptance correction to the reconstructed t distribution, using

$$c_f(t) = \frac{t_{\mathrm{MC}}}{t_{\mathrm{acc}}}, \tag{19}$$

where t_{MC} denotes the MC truth distribution and t_{acc} represents the distribution after applying detector acceptance and reconstruction effects.

The impact of the acceptance correction is illustrated on the right panels of Figures **??** and 20, which compares the reconstructed spectrum to the MC truth. Applying the correction factor restores the shape of the t distribution, enabling a meaningful comparison between reconstructed data and the underlying physics model.

6.2 Conversion to Differential Cross Section

Since the distribution is in counts, we must convert it into a differential cross section. This is done using Eq. 20

$$\frac{d\sigma}{dt} = \frac{1}{\Delta t \cdot L} \cdot \frac{N}{E_{\mathrm{ff}} \cdot A}, \tag{20}$$

where:

- Δt is the bin width of the momentum-transfer distribution,

- L is the integrated luminosity,
- N is the number of reconstructed events in a given t-bin,
- Eff is the detector efficiency,
- A is the acceptance correction factor.

7 Results

7.1 Exponential Fit to the t-Distribution

In exclusive scattering processes, one often performs a Fourier transform of the momentum-transfer distribution, $(d\sigma/dt)$, to move from momentum transfer space (t) to impact-parameter space (b_T), which characterizes the transverse spatial distribution of partons inside the proton.

The general form is

$$\tilde{f}(b_T) = \int dt\, e^{-ib_T t} f(t).$$

Fitting of ($d\sigma/dt$ in the right panel of Figure **??** gives Figure 21), where the t-distribution is described by an exponential ansatz

$$f(t) = Ae^{-Bt},$$

with fit parameters A and B, as shown in Figure 21

$$A = 0.0389 \pm 0.00018, \quad B = 3.85 \pm 0.013 \text{ GeV}^{-2}.$$

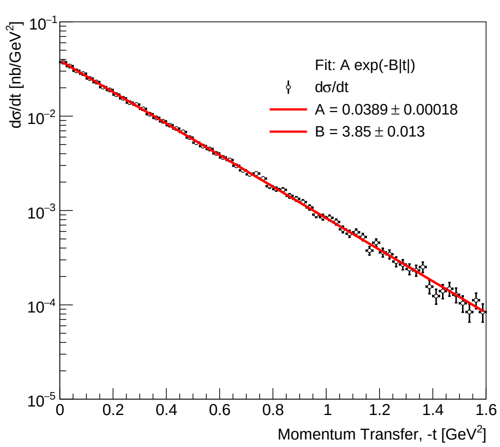

Figure 21: Determination of the t-slope parameter from the differential cross section. The differential cross section is fitted with the exponential function $d\sigma/dt = Ae^{-B|t|}$, yielding a slope parameter of $B = 3.85 \pm 0.013$ GeV^{-2}. The fitted function provides an excellent description of the simulated $-t$ dependence over the full kinematic range.

Thus, the Fourier transform expression of interest is given by:

$$\mathrm{Re}\Big(\tilde{f}(b_T)\Big) = \frac{A \cdot B}{B^2 + b_T^2}\left[(1 - e^{-(1.6B)}\cos(1.6b_T))\right] + \frac{A\, b_T}{B^2 + b_T^2}\left[e^{-(1.6B)}\sin(1.6b_T)\right]$$

See Appendix B for the Fourier formalism.

7.2 Gluon Radius Extraction

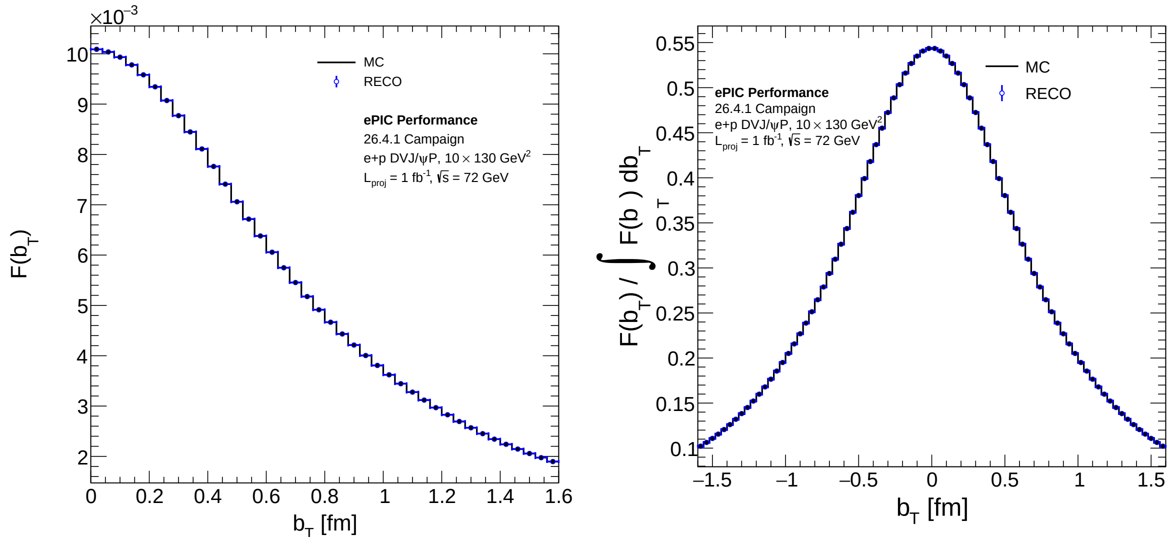

Figure 22: Fourier transform of the differential cross section, providing access to the spatial distribution of quarks within the nucleon.

From Figure 22, the Fourier transform of the measured differential cross section $d\sigma/dt$ provides quantitative access to the transverse spatial distribution of gluons in the proton. The extracted impact-parameter–space distribution $F(b_T)$ attains its maximum at $b_T = 0$ and decreases monotonically with increasing b_T, indicating a pronounced spatial localization of gluons in the proton core. The good agreement between the MC and corrected RECO results demonstrates that detector effects do not significantly bias the spatial imaging procedure. The width of the distribution is directly related to the t-slope parameter B, thereby enabling an estimate of the gluonic radius via

$$\langle b_T^2 \rangle \sim 2B.$$

These findings confirm the feasibility of gluon spatial imaging using exclusive J/ψ production at the EIC and open the way for systematic investigations of the proton's transverse spatial structure and small-x QCD dynamics.

8 Inclusive *ep* DIS Background

DIS represents a potential background to exclusive DVMP $J/\psi \to e^+e^-$ measurements when accidental combinations of final-state particles mimic the exclusive event topology. In particular, inclusive DIS events containing two oppositely charged leptons and a forward-going proton candidate may survive basic quality selections and enter the exclusive sample. A dedicated background study is therefore performed to evaluate the level of inclusive DIS impurity after applying the full set of exclusivity requirements.

The goals of the background study are to identify inclusive DIS events that mimic the exclusive $J/\psi(e^+e^-)$ final state, quantify the rejection power of the exclusivity selections, and assess any residual background contributions to the reconstructed t-distribution used in the cross-section determination.

8.1 Data Sample and Event Selection

The inclusive DIS background sample is obtained from simulated *ep* events generated with PYTHIA6 and reconstructed using the standard ePIC software framework. The sample corresponds to Early Science kinematics with $1 < Q^2 < 10$ GeV^2 and is processed through the same reconstruction and analysis chain as the DVMP signal sample to ensure a consistent comparison.

To emulate the exclusive J/ψ topology, the same baseline quality and exclusivity criteria used in the signal analysis are applied. Events are required to satisfy:

- exactly two oppositely charged tracks reconstructed in the central detector, consistent with an e^+e^- final state and within the central detector angular acceptance;
- a well-identified scattered electron satisfying the energy–momentum consistency requirement $0.9 < E/p < 1.2$ and standard DIS kinematic quality cuts;
- a J/ψ candidate formed from the e^+e^- pair with invariant mass within $2.9 < M_{e^+e^-} <$ 3.2 GeV;
- exactly one reconstructed positively charged particle in the far-forward region, identified as a recoil proton and detected in either the RP or the B0 spectrometer within their respective angular acceptances;
- a longitudinal energy-balance requirement to suppress photoproduction and radiative backgrounds;
- full reconstruction of all final-state particles together with a missing-mass requirement consistent with an exclusive final state.

PYTHIA-generated events containing genuine J/ψ production through the $J/\psi \to \mu^+\mu^-$ or $J/\psi \to e^+e^-$ decay channels are considered signal-like events. Therefore, they are removed from the DIS background sample so that signal events are not counted as background. Figure 23 shows that only two events from the PYTHIA DIS sample remain as background after the exclusive-like J/ψ events are removed. These remaining events are associated with misidentified kaons or pions. Since no particle-identification (PID) selection is applied at this stage, an improved rejection of the DIS background is expected once PID is implemented.

This requirement removes signal-like exclusive J/ψ contamination from the DIS background sample while retaining the non-exclusive DIS background. As shown in the left panel of Figure 23, the remaining DIS contribution is negligible and appears only in the first bin

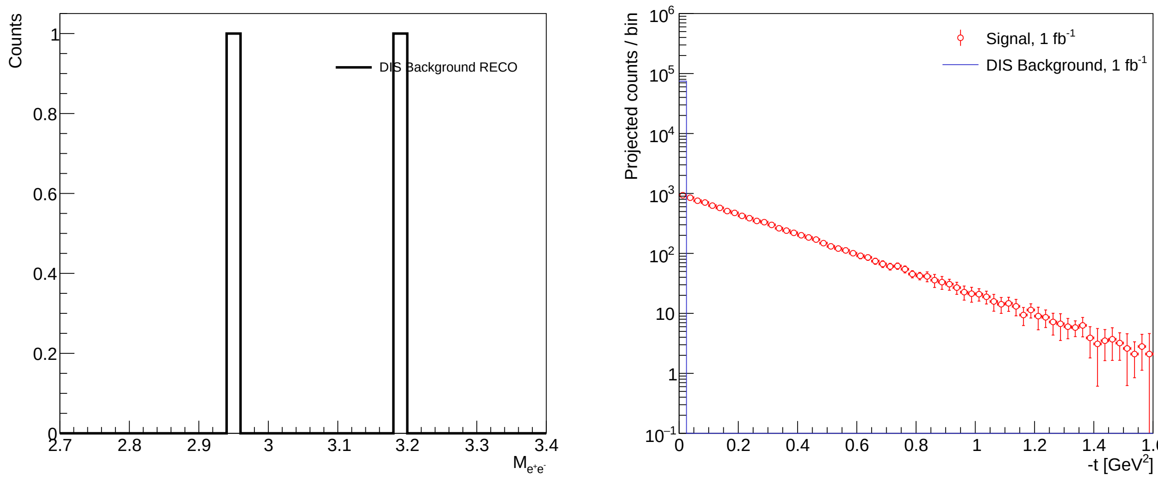

Figure 23: Comparison of reconstructed inclusive DIS background surviving the exclusive $J/\psi \to e^+e^-$ DVMP analysis selections after applying all quality and exclusivity cuts. The distributions show the reconstructed invariant mass (**Left**) and the projected event yields as a function of $-t$ for an integrated luminosity of 1 fb^{-1}. (**Right**), using the kinematic reconstruction methods discussed in Sections 5.2 and 5.3 and discussed in Section 5.4. The signal and DIS background yields are scaled to an integrated luminosity of 1 fb^{-1}.

of the projected $-t$ distribution shown in the right panel. The right panel of Figure 23 shows the projected event yields after applying efficiency corrections to the reconstructed signal. After this correction, the distribution becomes equivalent to the expected Monte Carlo yield with detector effects removed. The statistical uncertainties are obtained from the reconstructed event sample and propagated through the efficiency correction, resulting in the error bars shown on the corrected signal distribution.

The corrected projection therefore represents the expected measured yield after detector effects have been unfolded, while retaining the statistical precision determined by the reconstructed events. The projected DIS background is overlaid for comparison, demonstrating that the corrected signal remains well separated from the expected background over the full accessible $-t$ range.

8.2 Background Yield and Impact on the t-Distribution

After applying all quality and exclusivity selections to an inclusive DIS sample of 500,000 events, no events survive when the squared momentum transfer is reconstructed using **Method 1** ($t = (P_p - P_{p'})^2$) involving events from RP and B0 detectors. When the alternative kinematic reconstruction **Method 2** ($t_{\text{Method L}} = (P_p - P_{p'}^{\text{corr}})^2$), which does not require an explicitly reconstructed recoil proton, is used, a total of 2 events remain. These events are consistent with background fluctuations and slightly exhibit a resonant J/ψ signal structure in the reconstructed invariant mass distribution, as shown in the left panel of Figure 23.

The reconstructed t-distributions from the inclusive DIS sample, evaluated both prior to and following the application of luminosity-scaling corrections, as shown in the right panel of Figure 23, demonstrate that the residual contribution of inclusive DIS backgrounds to the final DVMP J/ψ measurement remains small and constrained to the lower t bin.

8.3 Systematic uncertainties

Several sources of systematic uncertainty are considered in this analysis. The uncertainties are evaluated independently in each $-t$ bin and combined in quadrature to obtain the total systematic uncertainty.

- **Luminosity:** A point-to-point correlated normalization uncertainty of 1.5% is assigned to account for the integrated luminosity determination, including beam-current normalization and detector live-time effects. This estimate is consistent with studies performed for ATHENA detector projections at the EIC and previous experience from HERA measurements [14, 15].

- **Event Selection:** The extracted cross-section is evaluated by varying the analysis selections around the nominal selection values. For example, the nominal invariant-mass selection, $2.9 < M_{J/\psi} < 3.2$ GeV, is varied to both a loose selection, $2.6 < M_{J/\psi} < 3.4$ GeV, and a tight selection, $3.05 < M_{J/\psi} < 3.15$ GeV. Similar variations are applied to the missing-mass requirement, proton acceptance cuts, and E/p selection.

  For each cut variation, the full analysis chain is repeated. In particular, the acceptance correction factor, defined as the inverse of the reconstruction efficiency, is recomputed and applied to obtain a corrected differential cross section. Although the reconstructed event yield slightly changes as the selections are loosened or tightened, the corrected cross section is expected to remain stable since the correction factor properly accounts for the variations.

  The systematic uncertainty associated with each selection is therefore determined from the residual variation of the corrected cross sections obtained from the loose, nominal, and tight cut configurations. The resulting bin-by-bin standard deviation is assigned as the systematic uncertainty associated with the event-selection procedure.

- **DIS Background:** Residual DIS contamination is estimated using the inclusive DIS background study described in Section 8.2. The corresponding background uncertainty is evaluated after scaling the surviving DIS background to the projected luminosity of the DVMP sample. The contribution to the uncertainty here comes from the **Method 2**. Statistical effects associated with the finite simulated DIS sample are then accounted for in order to extract the uncertainty on the residual background contamination.

9 Discussions

Following the validation of the reconstruction performance, a set of exclusivity requirements is imposed to isolate a high-purity sample of exclusive J/ψ events for the cross-section

Table 5: Summary of uncertainty sources. DIS value represent the statistical uncertainty after scaling to the projected luminosity value.

Source	Uncertainty Range (%)
Luminosity	1.5
Event selection	2.0–8.0
DIS background (Stat.)	0–1.0
Total (quadrature)	2.5–8.0

determination, as illustrated in Figure ??. The selected events are scaled to an integrated luminosity of $\mathcal{L} = 1\,\mathrm{fb}^{-1}$ per nucleon.

The squared four-momentum transfer, t, is reconstructed after the application of all quality and exclusivity selections. The resulting distributions are corrected for detector acceptance effects. Two complementary reconstruction approaches are employed. **Method 1** reconstructs t directly from the measured recoil proton momentum using the RP and B0 far-forward detector systems, while **Method 2** infers the recoil proton kinematics through the corrected missing-momentum formalism discussed in Section 5.3.

The reconstructed t-distributions obtained from the two methods are subsequently combined by prioritizing **Method 1** whenever direct proton reconstruction is available and using **Method 2** primarily in regions where **Method 1** is limited by detector acceptance gaps. This procedure provides continuous kinematic coverage over the accessible t-range in the present analysis[2]. The resulting t-distribution, shown in the right panel of Figure ??, is normalized by the integrated luminosity to obtain the differential cross section $d\sigma/dt$.

The extracted $d\sigma/dt$ distribution serves as the primary input for the Fourier-transform procedure used to access the transverse spatial distribution of gluons within the proton. The resulting impact-parameter distributions are presented in Figure 22.

The reconstructed t-distributions and Fourier-transform results demonstrate that the essential ingredients required for gluon imaging studies can be reproduced within a realistic detector simulation and reconstruction framework. Although the present study does not yet constitute a full projection analysis including complete systematic uncertainties and precision extraction of impact-parameter distributions, the results provide an important validation of the reconstruction methodology, detector performance, and analysis strategy required for future EIC gluon imaging measurements.

[2]Table 3 summarizes the detector angular acceptance and the corresponding t-range coverage.

References

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Appendices

A Additional Plots of Kinematic Variables

The distributions shown in Fig. 24 summarize the phase-space coverage and the performance of the t reconstruction. Panels (a) and (b) illustrate the Q^2–x_{bj} phase space at the generator and reconstructed levels, demonstrating that the detector acceptance preserves the overall kinematic reach of the process. Panels (c)–(e) present the t distributions and the correlation between reconstructed and true values, highlighting the limited acceptance of the RP and B0 detectors and the resulting gap in the intermediate $|t|$ region. This gap is mitigated in panel (g) by introducing the corrected momentum transfer t_{cor}, which combines information from both subsystems. The improved agreement between the reconstructed and MC distributions, as well as the smooth behavior of the cross section in panel (h), demonstrates the effectiveness of this approach in recovering the full t-spectrum.

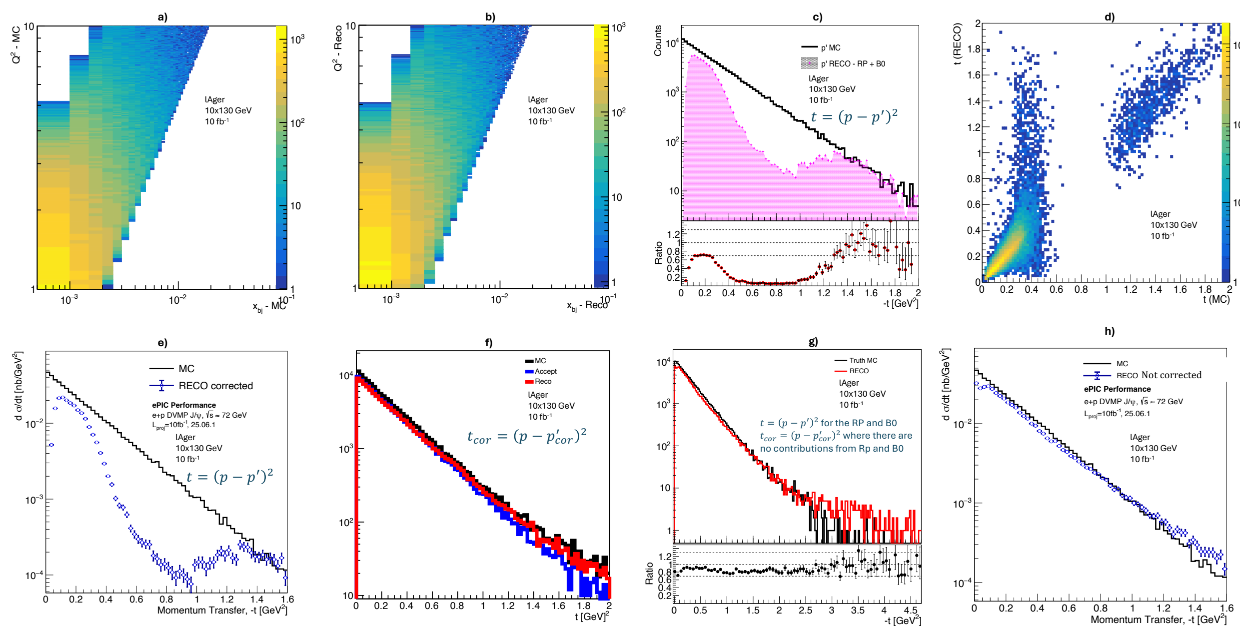

Figure 24: **a and b:** $Q^2 - x_{bj}$ phase space distributions; **c, d and e:** Transfer momentum distributions, correlation plot for the distribution form MC and RECO, and the cross section of the distribution; **f, g and h:** "f" employs a different method to reconstruct the transfer momentum, "g" uses t_{cor} to fill up the gap between RP and B0 where there are no detector proton events, and "h" gives the cross section distribution in "g".

Figure 25 shows the relative reconstruction resolution for x_{bj}, y, and Q^2 using the electron method. The distributions are generally centered around zero, indicating an overall unbiased reconstruction. However, a noticeable broadening is observed at low x_{bj} and low Q^2, reflecting the increased sensitivity of the electron method to detector resolution and other effects in this region. The resolution improves significantly toward higher values of the kinematic variables, where the scattered electron is better measured. These results highlight both the strengths and limitations of the electron method across the accessible phase space.

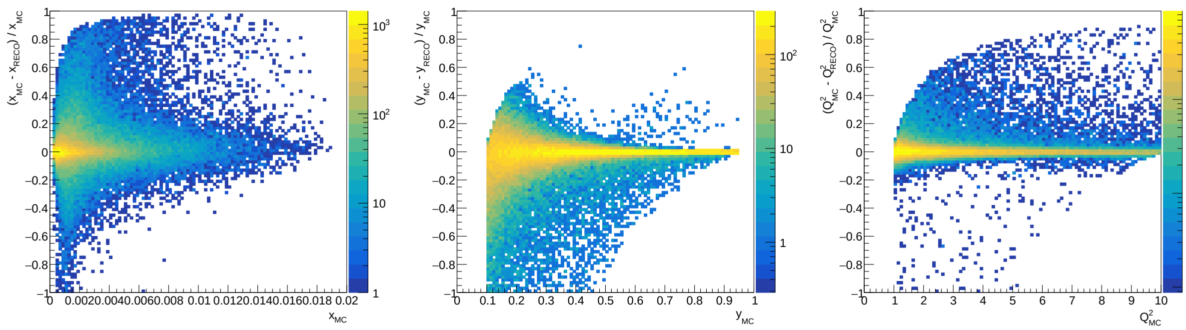

Figure 25: Relative reconstruction resolution for the kinematic variables x_{bj}, y, and Q^2 using the Electron method. Each panel shows the fractional difference $(X_{\mathrm{true}} - X_{\mathrm{reco}})/X_{\mathrm{true}}$ as a function of the true value of the corresponding variable, after applying all quality selections. The distributions illustrate the resolution and possible biases of the electron-based kinematic reconstruction.

The resolution obtained using the JB method is shown in Fig. 26. Compared to the electron method, the JB method exhibits a broader distribution in x_{bj}, particularly at low values, due to its reliance on the hadronic final state, which is more susceptible to detector acceptance and resolution effects. The y reconstruction shows improved stability in certain regions, while the Q^2 resolution remains comparable but slightly degraded relative to the electron method. These features reflect the complementary nature of the JB method, which is less sensitive to radiative effects but more dependent on hadronic reconstruction performance.

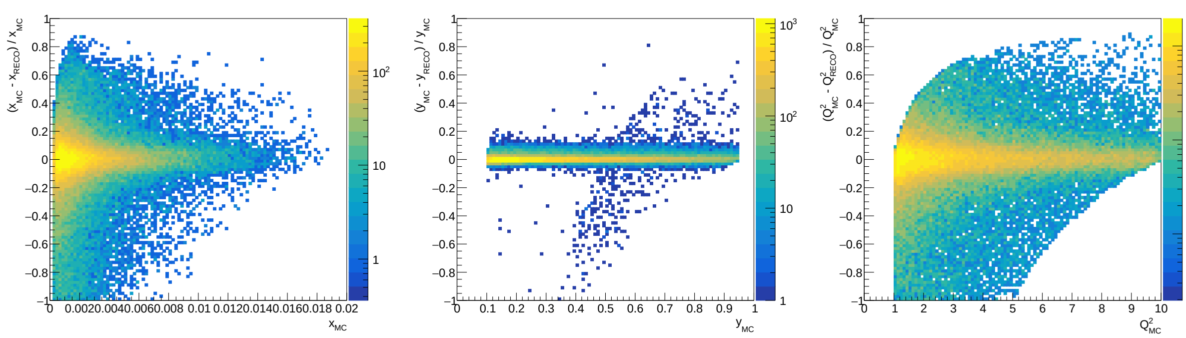

Figure 26: Relative reconstruction resolution for the kinematic variables x_{bj}, y, and Q^2 using the JB method. Each panel shows the fractional difference $(X_{\mathrm{true}} - X_{\mathrm{reco}})/X_{\mathrm{true}}$ as a function of the true value of the corresponding variable, after applying all quality selections.

Figure 27 presents the resolution for the $e\Sigma$ method, which combines information from both the scattered electron and the hadronic final state. The distributions show a clear improvement in resolution compared to the JB method and a more stable behavior across the full kinematic range relative to the electron method. In particular, the y and Q^2 reconstructions exhibit reduced spread and minimal bias, demonstrating the advantage of this hybrid

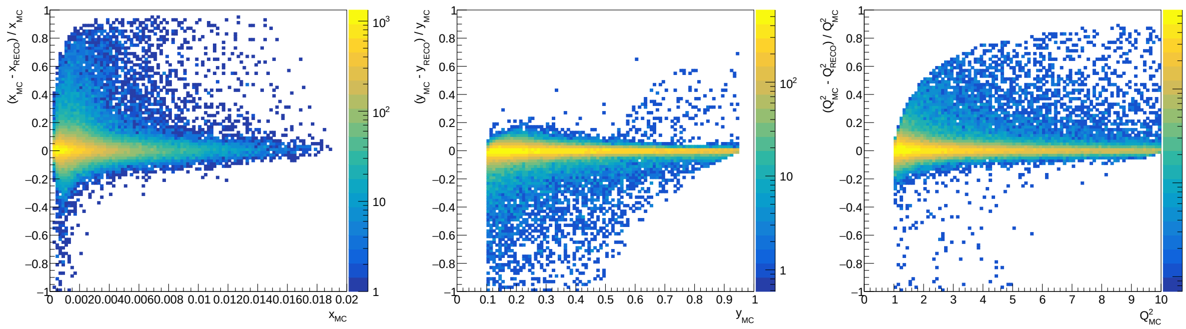

Figure 27: Relative reconstruction resolution for the kinematic variables x_{bj}, y, and Q^2 using the $e\Sigma$ method. Each panel shows the fractional difference $(X_{\mathrm{true}} - X_{\mathrm{reco}})/X_{\mathrm{true}}$ as a function of the true value of the corresponding variable, after applying all quality selections.

approach in balancing detector effects and radiative corrections. This confirms that the $e\Sigma$ method provides a robust and reliable reconstruction of DIS kinematics for the present analysis.

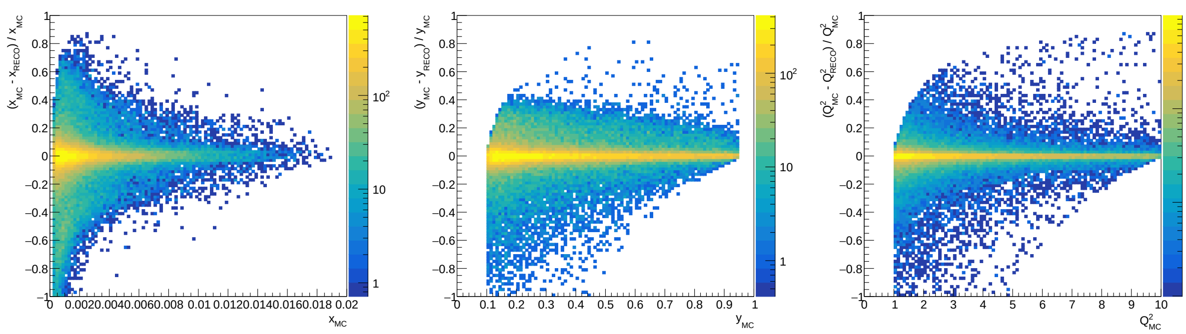

Figure 28: Relative reconstruction bias distributions for DA kinematic reconstruction method. The panels show the relative differences between reconstructed and MC values for x_{Bj} (left), Q^2 (middle), and y (right), defined as $(\mathrm{RECO} - \mathrm{MC})/\mathrm{MC}$, as a function of the corresponding MC variable.

In Figure 28, the narrow concentration of events around zero demonstrates the overall stability and resolution performance of the DA method across the studied kinematic range, while the larger spread at low x_{Bj} and low y reflects reduced reconstruction precision in regions with limited detector sensitivity and enhanced kinematic smearing.

Overall, the comparison of different reconstruction methods demonstrates that no single approach is optimal across the entire kinematic range. Instead, each method provides complementary strengths, motivating the use of multiple techniques for cross-checks and systematic uncertainty estimation in the final analysis.

B Further Fourier Transform Derivation

The general form is

$$\tilde{f}(b_T) = \int dt\, e^{-ib_T t} f(t).$$

where the t-distribution is described by an exponential ansatz

$$f(t) = Ae^{-Bt},$$

The fit parameters A and B for both MC and RECO are of great interest.

Thus, the Fourier transform reads

$$\tilde{f}(b_T) = \int dt\, e^{-ib_T t}\big(A\, e^{-Bt}\big) = A \int dt\, e^{-(B+ib_T)t}.$$

Evaluating the integral,

$$\tilde{f}(b_T) = -\frac{A}{B+ib_T}\left[e^{-(B+ib_T)t}\right]_{t_{min}}^{t_{max}}.$$

Where $|t_{min}| = 0$ and $|t_{max}| = 1.6$

$$\tilde{f}(b_T) = -\frac{A}{B+ib_T}\left[e^{-(1.6B+i1.6b_T)} - 1\right]$$

$$\tilde{f}(b_T) = -\frac{A}{B+ib_T}\left[e^{-(1.6B)}\cos(1.6b_T) - ie^{-(1.6B)}\sin(1.6b_T) - 1\right]$$

Multiplying the numerator and denominator by the complex conjugate,

$$\tilde{f}(b_T) = -\left\{\frac{A}{B+ib_T}\cdot\frac{B-ib_T}{B-ib_T}\right\}\left[e^{-(1.6B)}\cos(1.6b_T) - ie^{-(1.6B)}\sin(1.6b_T) - 1\right],$$

$$\tilde{f}(b_T) = -\left\{\frac{A\cdot B}{B^2+b_T^2} - i\frac{A\, b_T}{B^2+b_T^2}\right\}\left[e^{-(1.6B)}\cos(1.6b_T) - ie^{-(1.6B)}\sin(1.6b_T) - 1\right]$$

Thus, the Fourier transform separates into real and imaginary parts:

$$\mathrm{Re}\Big(\tilde{f}(b_T)\Big) = \frac{A\cdot B}{B^2+b_T^2}\left[(1 - e^{-(1.6B)}\cos(1.6b_T))\right] + \frac{A\, b_T}{B^2+b_T^2}\left[e^{-(1.6B)}\sin(1.6b_T)\right]$$

and

$$\mathrm{Im}\big(\tilde{f}(b_T)\big) = \frac{A\, b_T}{B^2+b_T^2}\left[e^{-(1.6B)}\cos(1.6b_T) - 1\right] + \frac{A\cdot B}{B^2+b_T^2}\left[e^{-(1.6B)}\sin(1.6b_T)\right]$$

Hence, for the phase such that $\tilde{f}(b_T) = |\tilde{f}(b_T)|e^{i\phi}$ we have;

$$\arg\Big(\tilde{f}(b_T)\Big) = \arctan\Big(\mathrm{Im}\Big(\tilde{f}(b_T)\Big), \mathrm{Re}\Big(\tilde{f}(b_T)\Big)\Big) \in (-\pi, \pi]$$

we note that $e^{-(1.6B)} << 1$, hence, by approximation we have;

$$\arg\big(\tilde{f}(b_T)\big) \approx -\arctan\Big(\frac{b_T}{B}\Big)$$

B.1 Detector Performance Across the (Q^2, x) Phase Space

Figures 29, 30, 31, and 32 present the reconstructed momentum-transfer $(-t)$ distributions in different (Q^2, x) bins for the three recoil-proton reconstruction approaches considered in this analysis: the RP, the B0 detector, and the corrected recoil reconstruction (**Method 2**). The reconstructed distributions are compared with the corresponding MC truth distribution to evaluate the detector acceptance and reconstruction performance throughout the accessible kinematic phase space.

Each figures corresponds to a fixed Q^2 interval, while the individual panels represent different Bjorken-x bins. This arrangement allows the detector performance to be examined over a broad range of DIS kinematics, from the lowest to the highest values of both Q^2 and x. Since the momentum transfer is directly related to the scattering angle and momentum of the recoil proton, the acceptance of the RP and B0 detectors varies significantly across the phase space. Consequently, the reconstruction efficiency depends not only on t but also on the underlying (Q^2, x) kinematics.

At low values of $-t$, the recoil proton is scattered through very small angles and remains close to the beam direction. In this region, the RP detector provides the highest reconstruction efficiency because it is specifically designed to detect forward-going protons with very small scattering angles. As the momentum transfer increases, the recoil proton is deflected to larger angles, gradually moving outside the RP acceptance. This behavior is clearly seen in nearly all (Q^2, x) bins, where the RP distribution decreases rapidly beyond approximately $-t \approx 0.3$ GeV^2.

The B0 detector complements the RP detector by extending the acceptance toward larger scattering angles and higher momentum transfers. Unlike the RP detector, the B0 detector contributes primarily in the high-t regions. The reconstructed B0 distributions become increasingly populated at larger values of $-t$, where the RP acceptance has already diminished. The overlap between the RP and B0 acceptance regions is due to smearing.

Between the RP and B0 acceptance regions, however, a transition region exists in which neither detector provides optimal efficiency. This reduction in acceptance is a direct consequence of the detector geometry and the beamline layout rather than a limitation of the reconstruction algorithms themselves. To recover events within this transition region, the corrected recoil reconstruction (**Method 2**) is employed. Method 2 reconstructs the recoil-proton kinematics indirectly using energy–momentum conservation together with the measured scattered electron and reconstructed J/ψ candidate. As observed throughout all four figures, **Method 2** successfully fills the regions where the direct proton detectors have reduced acceptance, thereby improving the continuity of the reconstructed $-t$ spectrum.

The contribution of **Method 2** is particularly evident at low momentum transfer, where direct proton reconstruction becomes increasingly difficult because the recoil proton remains extremely close to the beam axis. In addition, **Method 2** contributes significantly in the intermediate transition region between the RP and B0 detectors, recovering events that would otherwise be lost due to the limited geometric acceptance of the far-forward instrumentation. The reconstructed **Method 2** distributions generally follow the shape of the MC truth spectrum, demonstrating that the missing-momentum approach provides a reliable estimate of the recoil-proton kinematics in regions where direct detection is inefficient.

As the analysis progresses from the lowest Q^2 interval ($1 < Q^2 < 1.78$ GeV^2) to the highest interval ($5.62 < Q^2 < 10$ GeV^2), the overall event yield decreases considerably. This reduction reflects the rapidly falling exclusive J/ψ production cross section with increasing virtuality of the exchanged photon. Similarly, the event rate decreases toward larger Bjorken-x, leading to limited statistics in the highest-x bins. Consequently, several panels corresponding to large Q^2 and large x contain only a few reconstructed events or no reconstructed events at all. These empty bins are therefore a consequence of the limited event statistics rather than deficiencies in the reconstruction procedure.

Despite these statistical limitations, the relative detector behavior remains consistent across all four Q^2 intervals. The RP detector consistently dominates the low-t region, the B0 detector provides coverage at larger momentum transfers, and **Method 2** bridges the acceptance gap between the two direct proton detectors. This consistency demonstrates that the reconstruction strategy remains stable over the full kinematic region considered in this analysis.

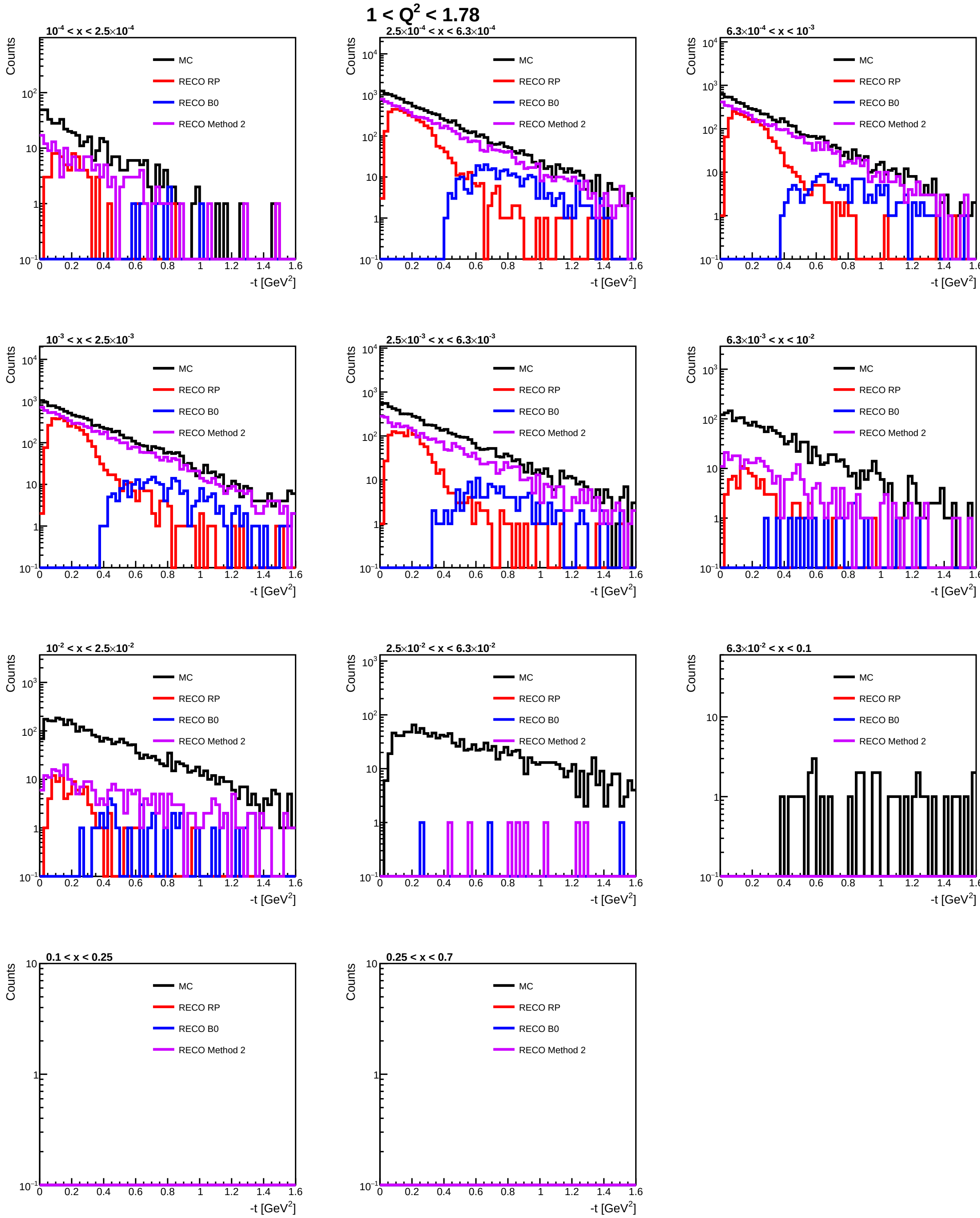

Figure 29: Comparison of the reconstructed $-t$ distributions obtained using the Roman Pot (RP), B0, and Method 2 with the corresponding MC truth distribution for $1 < Q^2 < 1.78$ GeV^2. Each panel represents a different Bjorken-x interval, illustrating the complementary detector acceptance and the contribution of Method 2 in recovering events within regions of reduced RP and B0 acceptance.

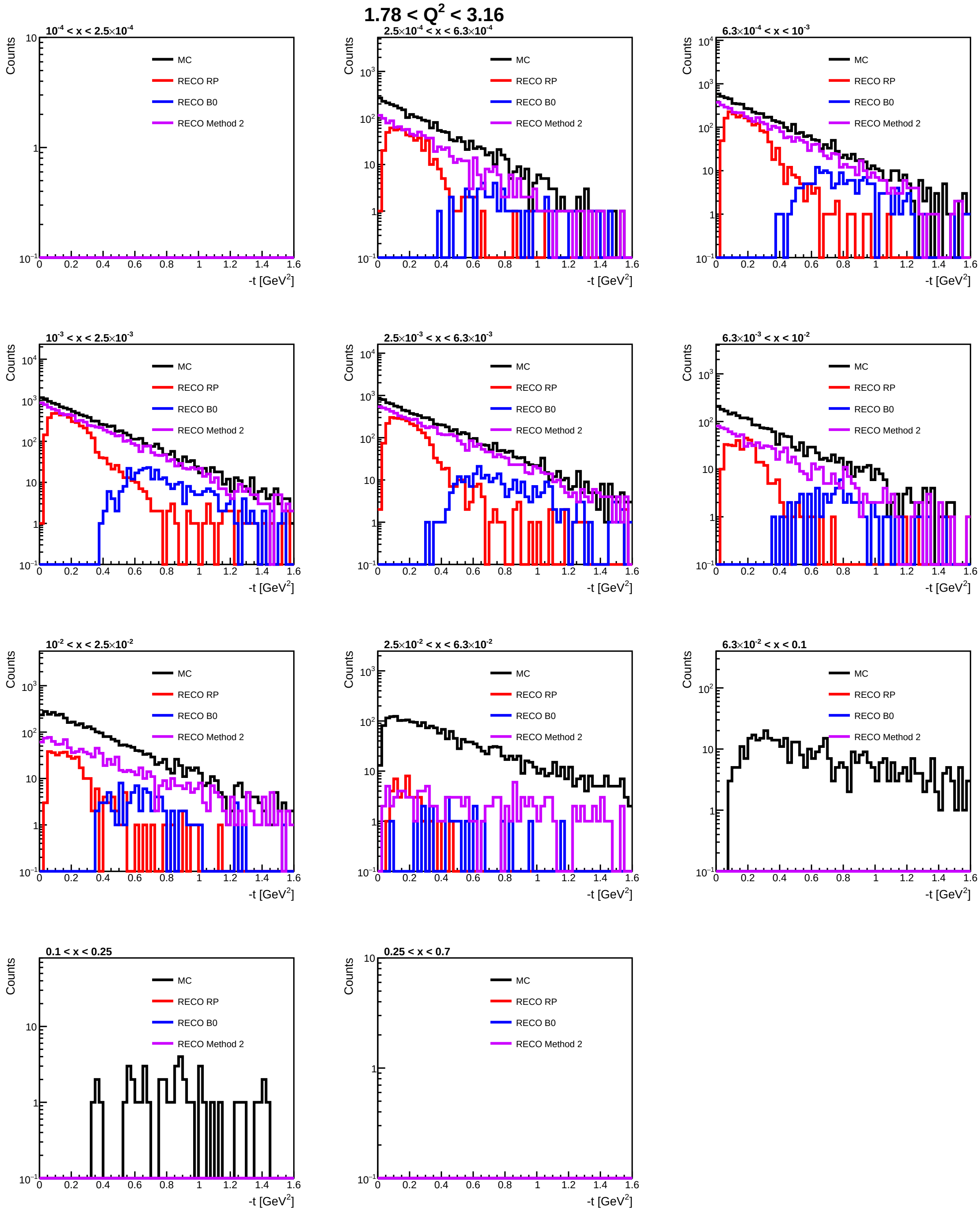

Figure 30: Comparison of the reconstructed $-t$ distributions obtained using the Roman Pot (RP), B0, and Method 2 with the corresponding MC truth distribution for $1.78 < Q^2 < 3.16$ GeV^2. Each panel represents a different Bjorken-x interval, illustrating the complementary detector acceptance and the contribution of Method 2 in recovering events within regions of reduced RP and B0 acceptance.

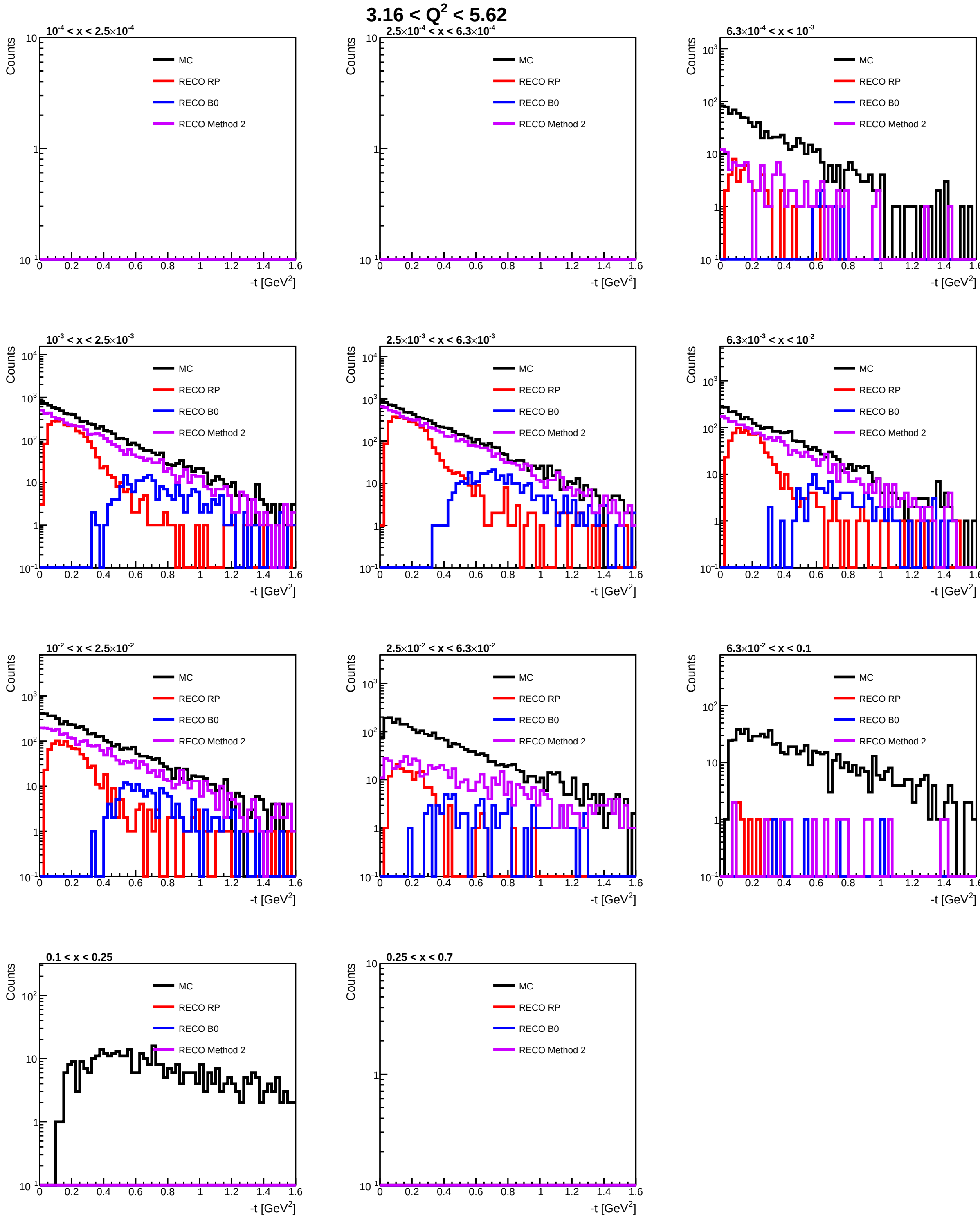

Figure 31: Comparison of the reconstructed $-t$ distributions obtained using the Roman Pot (RP), B0, and Method 2 with the corresponding MC truth distribution for $3.16 < Q^2 < 5.62$ GeV^2. Each panel represents a different Bjorken-x interval, illustrating the complementary detector acceptance and the contribution of Method 2 in recovering events within regions of reduced RP and B0 acceptance.

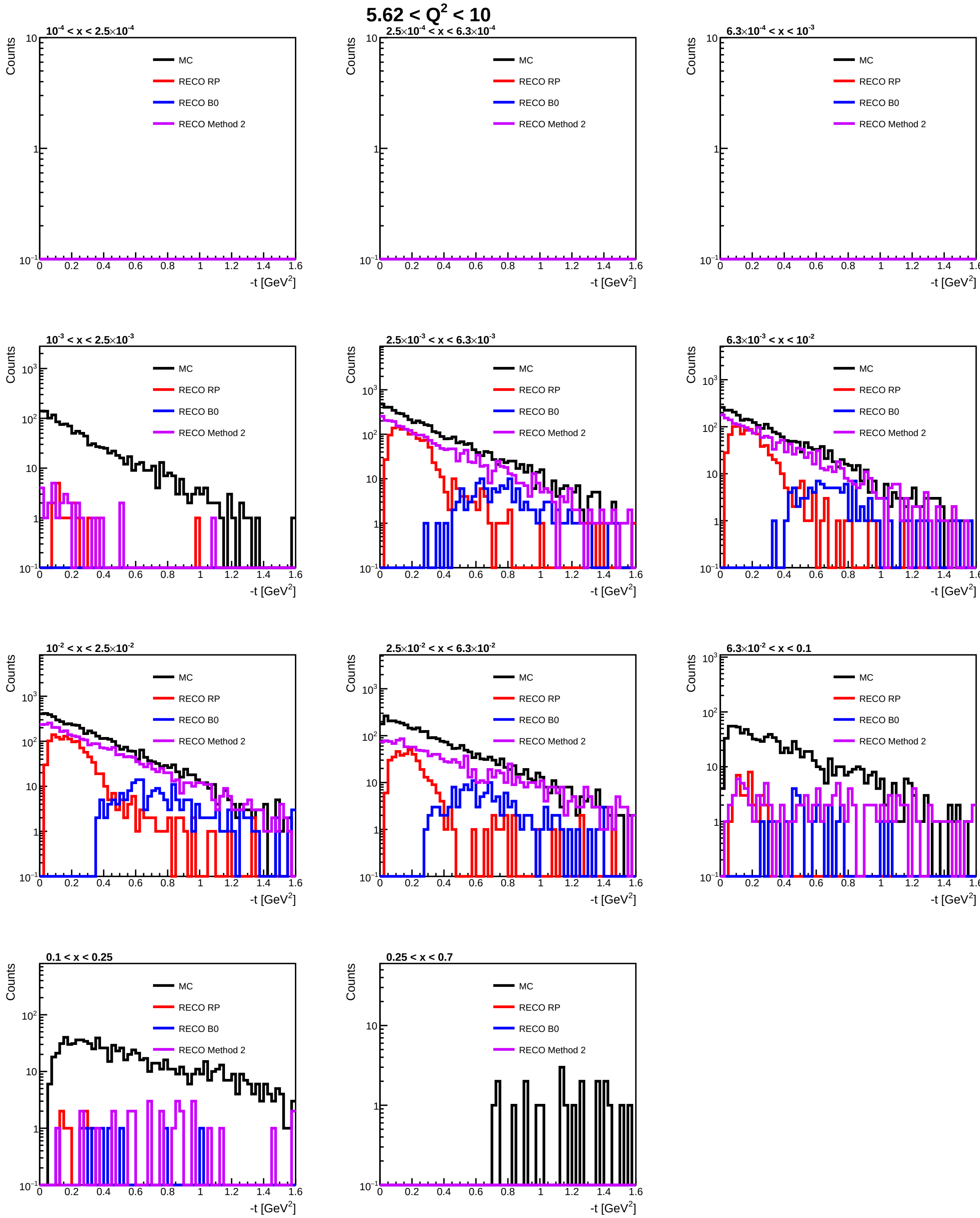

Figure 32: Comparison of the reconstructed $-t$ distributions obtained using the Roman Pot (RP), B0, and Method 2 with the corresponding MC truth distribution for $5.62 < Q^2 < 10$ GeV^2. Each panel represents a different Bjorken-x interval, illustrating the complementary detector acceptance and the contribution of Method 2 in recovering events within regions of reduced RP and B0 acceptance.